PROJECT REPORT ON

STATISTICAL ANALYSIS AND REPORTING OF CLINICAL TRIAL DATA USING SAS

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Project submitted for the award of the degree of

MASTER OF SCIENCE

BY

OSMANIA UNIVERSITY, HYDERABAD – 500007

HAINDAVI DEGREE & PG COLLEGE

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STATISTICAL ANALYSIS AND REPORTING OF CLINICAL TRIAL DATA USING SAS

This project work dissertation is submitted to

Osmania University, as partial fulfillment for the completion of M.Sc. III semester of

M.Sc.(Statistics)

By

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2025 - 2026

Date: 2026

DECLARATION

Thereby declare that the project work presented in this dissertation, entitled "**STATISTICAL ANALYSIS AND REPORTING OF CLINICAL TRIAL DATA USING SAS**" has been carried out by us under the supervision of Mrs. Vijayalaxmi, in the Department of Statistics, Haindavi Degree and PG College, Barkatpura, Hyderabad - 500027. The work done is Original and has not been submitted so far, in part or full, for any other degree or diploma to any University or Institution.

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CERTIFICATE

This is to certify that the project Report titled **STATISTICAL ANALYSIS AND REPORTING OF CLINICAL TRIAL DATA USING SAS** submitted in partial fulfillment for the award of **MSC (Statistics)** Programme of Department of Statistics, Haindavi Degree and PG College, Barkatpura, Hyderabad - 500027, was carried out by **PUJARI VEDAVATHI** H.T.NO:

1094-24-507-003, under my guidance. The work done is Original and has not been submitted so far, in part or full, for any other degree or diploma to any university or Institution.

Signature of Principal

Internal Examiner Signature with Date

External Examiner Signature with Date

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STATISTICAL ANALYSIS AND REPORTING OF CLINICAL TRIAL DATA USING SAS

ABSTRACT

Clinical trials are essential for evaluating the safety and efficacy of new medical treatments before regulatory approval. Statistical analysis plays a critical role in ensuring that clinical trial data are accurately interpreted and reported according to regulatory standards. The present study focuses on the statistical analysis and reporting of clinical trial data using SAS software.

The primary objective of this project is to analyze demographic and baseline characteristics of patients enrolled in a randomized clinical trial and to construct regulatory-style summary tables. The dataset used in this study consists of patient-level information including treatment group, date of birth, diagnosis date, age, gender, and race.

Descriptive statistical methods were applied to summarize continuous variables, including **mean, standard deviation, minimum, and maximum** values. Categorical variables such as gender and race were analyzed using frequency distributions and percentage calculations. Statistical procedures such as PROC MEANS and PROC FREQ were implemented in SAS to generate summary statistics. Data manipulation techniques including **sorting, formatting, transposing, and reporting** were used to construct structured demographic tables comparable to standard clinical trial reporting formats.

This study demonstrates the application of statistical techniques in generating Table 1: Demographic and Baseline Characteristics by Treatment Group, an essential component of clinical trial reporting. Inferential statistical methods including independent **sample t-test, one-way ANOVA, and Chi-square** tests were performed to assess comparability between treatment groups. The results showed no statistically significant differences in demographic characteristics across groups ($p > 0.05$), confirming baseline balance prior to further analysis. The project emphasizes the importance of data preprocessing, statistical testing, and standardized reporting using SAS in supporting evidence-based clinical research.

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CHAPTER 1

INTRODUCTION

1.1 Overview of Clinical Trials

Clinical trials are systematic and scientifically controlled studies conducted to evaluate the safety, efficacy, and effectiveness of medical interventions such as drugs, vaccines, medical devices, or treatment procedures. They form the backbone of evidence-based medicine and are essential before any new therapeutic product is approved for public use.

1.2 Phases of Clinical Trials

Phase I

- Conducted in a small group of participants (usually healthy volunteers or patients).
- Primary focus: **Safety and tolerability**.
- Determines safe dosage range.

Phase II

- Conducted in a larger group of patients with the target condition.
- Primary focus: **Preliminary efficacy**.
- Continues safety evaluation.

Phase III

- Conducted in large populations.
- Confirms **treatment efficacy**.
- Compares new treatment with standard therapy or placebo.

Phase IV

- Conducted after regulatory approval.
- Evaluates **long-term safety and effectiveness**.
- Assesses real-world performance.

The success of a clinical trial depends not only on medical design but also on rigorous statistical planning and analysis. Proper statistical methodology ensures that results are scientifically valid, unbiased, and reproducible.

1.3 Role of Biostatistics in Clinical Research

Biostatistics plays a critical role in all stages of a clinical trial, from design to reporting. It helps in determining appropriate sample size, selecting suitable statistical tests, defining endpoints, and ensuring proper randomization.

Key contributions of biostatistics include:

- Designing randomized controlled trials
- Estimating treatment effects
- Measuring variability in patient responses
- Testing statistical hypotheses
- Controlling Type I and Type II errors
- Constructing confidence intervals
- Ensuring regulatory compliance

Statistical methods help distinguish between true treatment effects and random variation. Without statistical analysis, clinical data cannot be interpreted meaningfully.

1.4 Importance of Statistical Reporting in Clinical Trials

Statistical reporting in clinical trials follows strict regulatory guidelines. Regulatory authorities require standardized presentation of results to ensure transparency and scientific integrity.

One of the most important components of clinical trial reporting is the presentation of demographic and baseline characteristics. This information confirms whether treatment groups are comparable before evaluating treatment efficacy.

Descriptive statistics are used to summarize patient characteristics, while inferential statistics are used to test for significant differences between treatment groups. Proper reporting includes tables, graphical summaries, hypothesis testing results, and interpretation of findings.

Statistical reporting ensures that conclusions drawn from clinical trials are based on objective evidence rather than subjective judgment.

1.5 Need for the Study

With increasing complexity in clinical research and growing regulatory requirements, there is a demand for professionals skilled in clinical data analysis and statistical reporting.

The study demonstrates:

- Data preprocessing techniques
- Derivation of new variables (such as age calculation)
- Application of descriptive and inferential statistical methods
- Construction of structured summary tables
- Interpretation of statistical results

By implementing these techniques using SAS software, the study reflects industry-relevant methodologies used in pharmaceutical and clinical research organizations.

1.6 Organization of the Report

The report is organized into multiple chapters covering literature review, objectives, methodology, statistical techniques, data analysis, discussion, and conclusions. Each chapter systematically presents theoretical concepts and practical implementation related to clinical trial data analysis and reporting.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The literature review provides a comprehensive understanding of existing research and methodologies related to statistical analysis in clinical trials. Clinical research relies heavily on statistical principles to ensure validity, reliability, and reproducibility of findings. Numerous studies have emphasized the importance of proper statistical planning, data analysis, and structured reporting in randomized controlled trials.

This chapter reviews previous research on statistical methodologies used in clinical trials, reporting standards, baseline characteristic analysis, and the application of statistical software in clinical data management.

2.2 Statistical Methods in Clinical Trials

Statistical methodology forms the foundation of clinical trial design and analysis. Randomized controlled trials (RCTs) are widely recognized as the gold standard in clinical research because they reduce bias and improve comparability between treatment groups.

Research highlights the importance of:

- Randomization techniques
- Sample size estimation
- Hypothesis testing
- Confidence interval estimation
- Handling missing data

Descriptive statistics are used to summarize patient demographics and baseline characteristics. Inferential statistics are used to determine whether observed differences between treatment groups are statistically significant.

Studies have shown that improper statistical analysis can lead to misleading conclusions, which may affect regulatory decisions and patient safety.

2.3 Baseline Characteristics Reporting

Baseline characteristics are essential in evaluating whether treatment groups are comparable at the start of a clinical trial. Literature emphasizes that demographic variables such as age, gender, and race must be summarized clearly and accurately.

Regulatory guidelines recommend presenting baseline data in a structured tabular format, commonly referred to as "Table 1". This table typically includes:

- Mean and standard deviation for continuous variables
- Frequency and percentage for categorical variables

Researchers stress that balanced baseline characteristics reduce confounding effects and increase the credibility of trial results.

2.4 Hypothesis Testing in Clinical Research

Hypothesis testing is a fundamental statistical tool used in clinical trials. The null hypothesis generally states that there is no difference between treatment groups, while the alternative hypothesis suggests a difference exists.

Commonly used tests in clinical trials include:

- Student’s t-test for comparing means
- Chi-square test for categorical variables
- Analysis of Variance (ANOVA)
- Non-parametric tests when normality assumptions are violated

Literature indicates that proper interpretation of p-values and confidence intervals is essential for making valid clinical conclusions. Overreliance on p-values without clinical interpretation may lead to incorrect decision-making.

2.5 Role of Statistical Software in Clinical Trials

Modern clinical trials depend on statistical software for data management, analysis, and reporting. Software such as SAS is widely used in pharmaceutical industries and contract research organizations (CROs) due to its reliability and regulatory acceptance.

Research articles highlight the importance of:

- Data validation
- Variable derivation
- Generation of tables, listings, and figures (TLFs)
- Reproducible analysis

The use of structured programming ensures transparency and consistency in clinical trial reporting.

2.6 Gaps Identified in Literature

Although many studies focus on advanced statistical models, fewer studies emphasize practical implementation of statistical reporting using industry-standard software tools. There is a need for integrated projects that combine statistical theory with practical SAS-based reporting methods.

This study aims to bridge that gap by demonstrating statistical analysis and reporting of clinical trial data using structured SAS programming techniques.

2.7 Summary of Literature Review

The literature review establishes that statistical analysis is central to clinical trial research. Proper baseline reporting, hypothesis testing, and regulatory-compliant data presentation are essential components of clinical data analysis. The reviewed studies highlight the importance of combining statistical theory with practical implementation to ensure reliable and interpretable results.

CHAPTER 3

OBJECTIVES OF THE STUDY

Introduction

The primary purpose of this study is to demonstrate the application of statistical techniques in the analysis and reporting of clinical trial data. The study focuses on structured statistical evaluation of demographic and baseline characteristics using SAS software. The objectives are formulated to ensure systematic analysis, appropriate hypothesis testing, and regulatory-compliant reporting of results.

3.1 Primary Objective

The primary objective of this study is:

- To perform statistical analysis and reporting of clinical trial demographic and baseline data using SAS programming techniques.

3.2 Secondary Objectives

The secondary objectives of the study include:

1. To derive necessary variables such as age from available date variables.
2. To summarize continuous variables using descriptive statistics such as mean, standard deviation, minimum, and maximum.
3. To summarize categorical variables using frequency distributions and percentage calculations.
4. To compare treatment groups using appropriate statistical hypothesis tests.
5. To construct structured demographic tables similar to standard clinical trial reporting formats.
6. To generate graphical representations for better visualization of data distribution.
7. To interpret statistical results in the context of clinical research.

3.3 Scope of the Study

The scope of this study is limited to:

- Analysis of demographic and baseline variables.
- Application of descriptive and inferential statistical methods.
- Construction of structured summary tables and graphical outputs.
- Implementation using SAS software.

The study does not include advanced predictive modeling or machine learning techniques, as the focus is on structured clinical trial statistical reporting.

Significance of the Study

This study is significant because:

- It demonstrates practical implementation of statistical theory in clinical trial analysis.
- It aligns with industry standards used in pharmaceutical and clinical research organizations.
- It enhances understanding of regulatory-compliant statistical reporting.
- It provides a structured approach to analyzing clinical trial data.

CHAPTER 4

SOURCE OF DATA AND DATA DESCRIPTION

Introduction

This chapter presents the source of data used in the study, the structure of the dataset, description of variables, and preprocessing steps performed prior to statistical analysis. Understanding the nature and structure of the dataset is essential for applying appropriate statistical techniques in clinical trial analysis.

4.1 Source of Data

The dataset used in this study was obtained as part of a Clinical Research / SAS training module conducted through the Internshala online learning platform. The dataset was provided for academic and training purposes to demonstrate clinical trial data handling and statistical reporting techniques.

The dataset represents a simulated randomized clinical trial structure designed to resemble real-world clinical research data. It contains subject-level demographic information and treatment group allocation necessary for performing descriptive and inferential statistical analysis.

The dataset is used strictly for academic analysis and does not contain identifiable personal health information.

4.2 Nature of the Study Dataset

The dataset mimics a randomized clinical trial with multiple treatment arms. Each subject is uniquely identified and assigned to a treatment group. The dataset structure follows typical clinical trial data conventions used in pharmaceutical research.

The primary focus of the dataset includes:

- Treatment allocation
- Demographic variables
- Date variables required for age derivation
- Categorical variables for baseline comparison

4.3 Study Population

The study population consists of subjects randomized into two treatment groups:

- Treatment Group 0 – Placebo
- Treatment Group 1 – Active Treatment

For reporting purposes, an additional derived group was created:

- Treatment Group 2 – Total Population

This derived total population facilitates presentation of combined statistics in summary tables.

4.4 Data Structure

The dataset is organized in a tabular format where:

- Each row represents one subject
- Each column represents a variable

The dataset includes the following key variables:

Variable Name	Description
STUDY	Study identifier
PATNO	Patient number
SITENO	Site number
SUBJINI	Subject initials
TRT	Treatment group indicator
DAY	Day of birth
MONTH	Month of birth
YEAR	Year of birth
DIAGDT	Diagnosis date
GENDER	Gender code
RACE	Race code

Variable Description

Treatment Variable (TRT)

The treatment variable indicates assignment of subjects to treatment arms:

- 0 = Placebo
- 1 = Active Treatment
- 2 = Total Population (Derived for reporting)

Continuous Variable

The primary continuous variable analyzed in this study is:

- Age (derived variable)

Age was calculated based on date of birth and diagnosis date.

Categorical Variables

The categorical variables analyzed include:

- Gender (Male / Female)
- Race (White, African American, Hispanic, Asian, Other)

These variables were summarized using frequency distributions and percentage calculations.

4.5 Data Preprocessing

Prior to statistical analysis, several preprocessing steps were performed.

Construction of Date of Birth

The date of birth was derived by combining day, month, and year variables. The combined character date was converted into SAS date format for accurate calculation.

Derivation of Age

Age was calculated using the formula:

Age = (Diagnosis Date − Date of Birth) / 365.25

This ensures accurate age estimation while accounting for leap years.

Creation of Total Treatment Group

To generate comprehensive summary tables including overall population statistics, the dataset was duplicated and assigned a new treatment group value representing the total population.

Data Validation and Quality Checks

Data validation steps were performed to ensure:

- Proper date formatting
- Accurate numeric conversion
- Correct treatment group assignment
- Absence of invalid numeric data

These validation procedures are essential to avoid computational errors and ensure reliability of statistical outputs.

Summary

This chapter described the source, structure, and characteristics of the dataset used in the study. The data obtained from the Internshala training module were structured in a format consistent with clinical trial datasets. Appropriate preprocessing and derivation steps were performed prior to statistical analysis.

The next chapter presents the methodology adopted for analyzing and reporting the clinical trial data using statistical techniques and SAS programming.

CHAPTER 5

METHODOLOGY

Introduction

This chapter describes the statistical methodology and analytical procedures adopted for the clinical trial data analysis. The analysis was performed using SAS software following standard clinical trial reporting practices. The methodology includes data preparation, derivation of variables, descriptive statistics, inferential analysis, and structured statistical reporting.

Study Design

The dataset represents a randomized parallel-group clinical trial design. Subjects were allocated into two treatment groups:

- Treatment Group 0 – Placebo
- Treatment Group 1 – Active Treatment

For reporting purposes, a derived total population group (Treatment = 2) was created.

The study design supports comparison of demographic characteristics across treatment groups.

5.1 Data Preprocessing

Description

Data preprocessing involves transforming raw data into a structured format suitable for statistical analysis. In this study, preprocessing included creation of date variables and preparation of dataset for analysis.

Before statistical analysis, the dataset underwent the following preparation steps:

5.1 Creation of Date of Birth

The date of birth was constructed by combining:

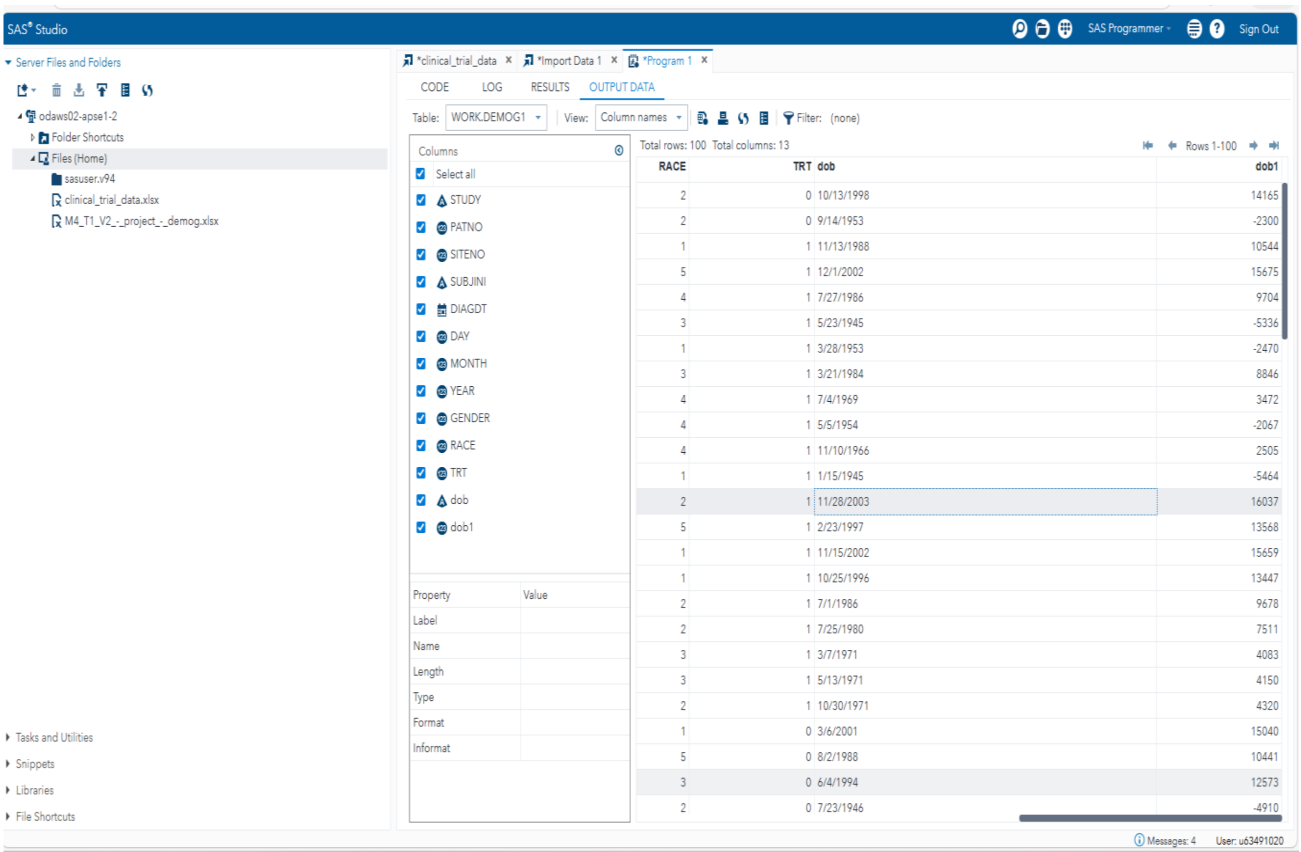
- DAY
- MONTH
- YEAR

The combined character variable was converted into SAS date format using the INPUT function.

SAS Code:

```
data demog1;
 set demog;
 dob=compress(cat(month,'/',day,'/',year));
 dob1=input(dob,mmddyy10.);
run;
```

Output:



Interpretation:

In this step, a new dataset demog1 is created from the original dataset demog. The variables month, day, and year are combined to form a complete date of birth in character format. Then, this character date is converted into a numeric SAS date variable using the INPUT function with MMDDYY10.

5.2 Data Cleaning Techniques

Description

Data cleaning ensures that variables are correctly formatted and free from inconsistencies before analysis.

In this study, formatting and validation checks were performed.

SAS Code:

```
proc contents data=demog1;
run;
proc means data=demog1 n nmiss;
 var dob1 diagdt1;
run;
```

Output:

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat	Label
6	DAY	Num	8	BEST.		DAY
5	DIAGDT	Num	8	MMDDYY10.		DIAGDT
9	GENDER	Num	8	BEST.		GENDER
7	MONTH	Num	8	15.		MONTH
2	PATNO	Num	8	BEST.		PATNO
10	RACE	Num	8	BEST.		RACE
3	SITENO	Num	8	BEST.		SITENO
1	STUDY	Char	7	$7.	$7.	STUDY
4	SUBJINI	Char	4	$4.	$4.	SUBJINI
11	TRT	Num	8	BEST.		TRT
8	YEAR	Num	8	BEST.		YEAR
15	age	Num	8			
14	diagdt1	Num	8	DATE9.		
12	dob	Char	200			
13	dob1	Num	8	DATE9.		

Sort Information

Sortedby	TRT
Validated	YES
Character Set	ASCII

The MEANS Procedure

Variable	N	N Miss
dob1	200	0
diagdt1	200	0

Interpretation

The PROC CONTENTS output provides details regarding variable names, types, lengths, and formats present in the dataset.

From the output:

- The dataset contains both numeric and character variables.
- STUDY and SUBJINI are character variables.
- Clinical and demographic variables such as PATNO, SITENO, TRT, GENDER, and RACE are numeric.
- The derived variables dob1 and diagdt1 are stored as numeric variables with DATE9. format.
- The derived variable age is stored as a numeric variable.

The presence of appropriate date formats (DATE9.) confirms that date variables are correctly structured for performing date calculations such as age derivation.

This verifies that the dataset structure is appropriate for statistical analysis.

The PROC MEANS procedure was used to check for missing values in the derived date variables (dob1 and diagdt1).

From the output:

- Total number of observations (N) = 100
- Number of missing values (N Miss) = 0

for both dob1 and diagdt1.

This indicates that:

- All 100 subjects have valid Date of Birth values.
- All 100 subjects have valid Diagnosis Date values.
- There are no missing date entries in the dataset.

The absence of missing values ensures completeness of the dataset and confirms that age calculation and subsequent statistical analysis can be performed without data-related errors.

5.3 Variable Derivation Methods

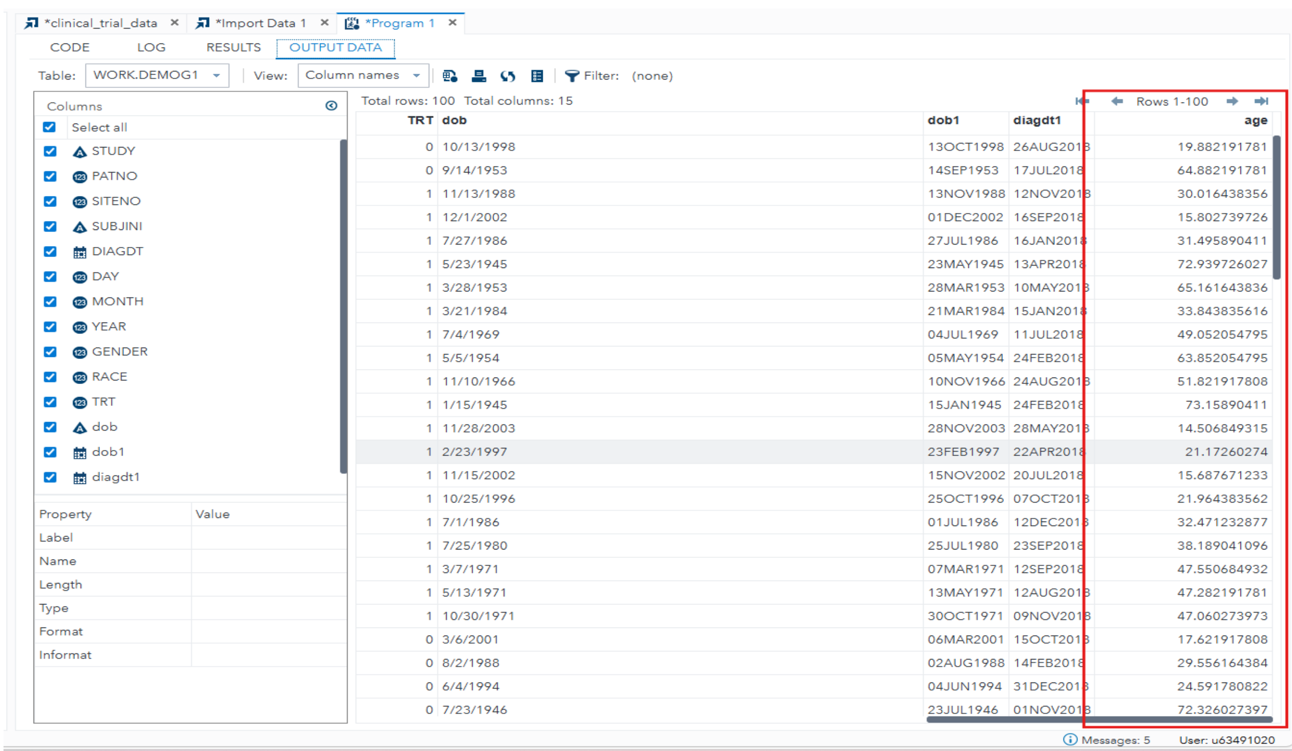
Deriving the age variable:

Age= (Diagnosis Date−Date of Birth)/365

SAS Code:

```
data demog1;
 set demog;
 dob=compress(cat(month,'/',day,'/',year));
 dob1=input(dob,mmddyy10.);
 format dob1 date9.;
 diagdt1 = diagdt;
 format diagdt1 date9.;
 age=(diagdt1-dob1)/365;
run;
```

Output:



TRT	dob	dob1	diagdt1	age
0	10/13/1998	13OCT1998	26AUG2018	19.882191781
0	9/14/1953	14SEP1953	17JUL2018	64.882191781
1	11/13/1988	13NOV1988	12NOV2018	30.016438356
1	12/1/2002	01DEC2002	16SEP2018	15.802739726
1	7/27/1986	27JUL1986	16JAN2018	31.495890411
1	5/23/1945	23MAY1945	13APR2018	72.939726027
1	3/28/1953	28MAR1953	10MAY2018	65.161643836
1	3/21/1984	21MAR1984	15JAN2018	33.843835616
1	7/4/1969	04JUL1969	11JUL2018	49.052054795
1	5/5/1954	05MAY1954	24FEB2018	63.852054795
1	11/10/1966	10NOV1966	24AUG2018	51.821917808
1	1/15/1945	15JAN1945	24FEB2018	73.15890411
1	11/28/2003	28NOV2003	28MAY2018	14.506849315
1	2/23/1997	23FEB1997	22APR2018	21.17260274
1	11/15/2002	15NOV2002	20JUL2018	15.687671233
1	10/25/1996	25OCT1996	07OCT2018	21.964383562
1	7/1/1986	01JUL1986	12DEC2018	32.471232877
1	7/25/1980	25JUL1980	23SEP2018	38.189041096
1	3/7/1971	07MAR1971	12SEP2018	47.550684932
1	5/13/1971	13MAY1971	12AUG2018	47.282191781
1	10/30/1971	30OCT1971	09NOV2018	47.060273973
0	3/6/2001	06MAR2001	15OCT2018	17.621917808
0	8/2/1988	02AUG1988	14FEB2018	29.556164384
0	6/4/1994	04JUN1994	31DEC2018	24.591780822
0	7/23/1946	23JUL1946	01NOV2018	72.326027397

Deriving the agegroup variable:

Description:

In this step, a new categorical variable agegroup was created from the continuous variable age using a DATA step. The dataset demog1 was read, and a new dataset demog4 was generated with the derived variable.

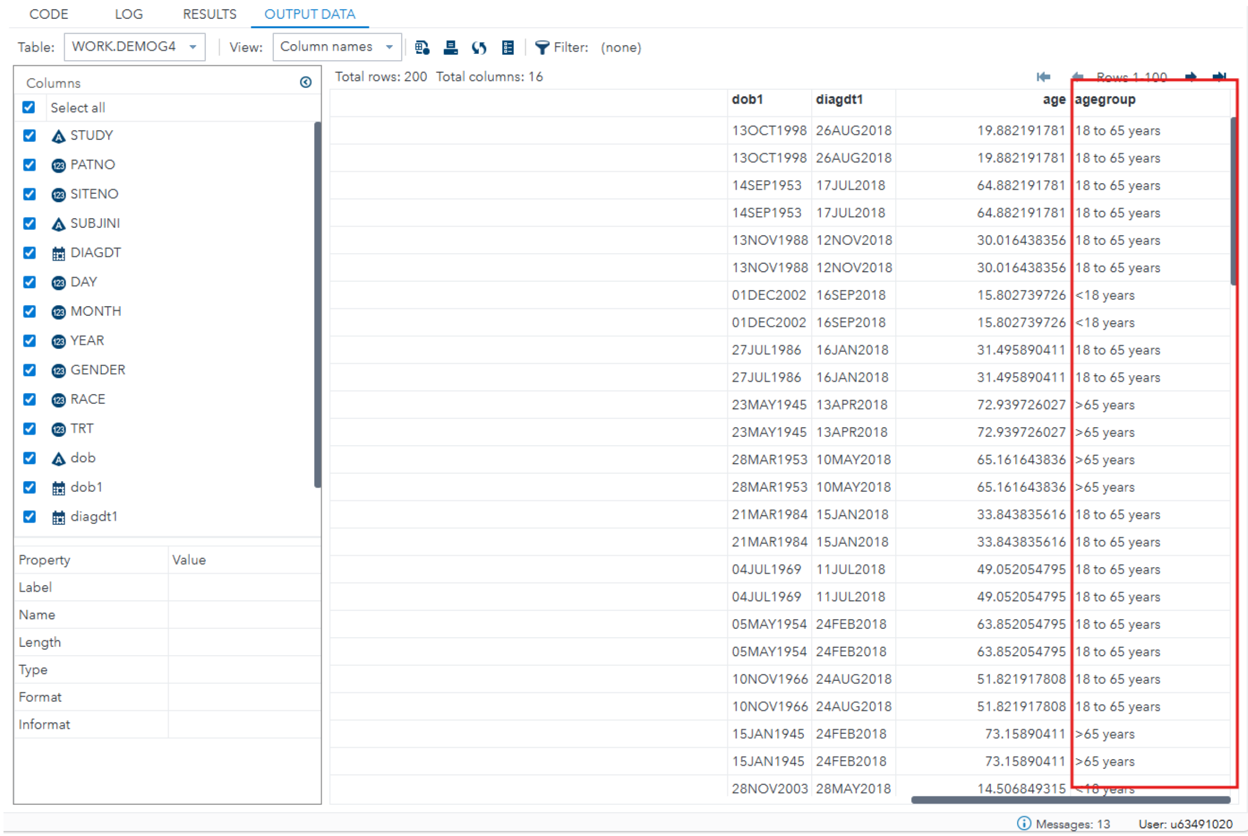
The LENGTH statement defined agegroup as a character variable with sufficient space to store category labels. The IF–ELSE conditions categorized patients into three groups:

- <18 years for subjects younger than 18
- 18 to 65 years for subjects between 18 and 65
- >65 years for subjects older than 65

SAS Code:

```
data demog4;
   set demog1;
   length agegroup $20;
   if age < 18 then agegroup = '<18 years';
   else if 18 <= age <= 65 then agegroup = '18 to 65 years';
   else if age > 65 then agegroup = '>65 years';
run;
```

Output:



dob1	diagdt1	age	agegroup
13OCT1998	26AUG2018	19.882191781	18 to 65 years
13OCT1998	26AUG2018	19.882191781	18 to 65 years
14SEP1953	17JUL2018	64.882191781	18 to 65 years
14SEP1953	17JUL2018	64.882191781	18 to 65 years
13NOV1988	12NOV2018	30.016438356	18 to 65 years
13NOV1988	12NOV2018	30.016438356	18 to 65 years
01DEC2002	16SEP2018	15.802739726	<18 years
01DEC2002	16SEP2018	15.802739726	<18 years
27JUL1986	16JAN2018	31.495890411	18 to 65 years
27JUL1986	16JAN2018	31.495890411	18 to 65 years
23MAY1945	13APR2018	72.939726027	>65 years
23MAY1945	13APR2018	72.939726027	>65 years
28MAR1953	10MAY2018	65.161643836	>65 years
28MAR1953	10MAY2018	65.161643836	>65 years
21MAR1984	15JAN2018	33.843835616	18 to 65 years
21MAR1984	15JAN2018	33.843835616	18 to 65 years
04JUL1969	11JUL2018	49.052054795	18 to 65 years
04JUL1969	11JUL2018	49.052054795	18 to 65 years
05MAY1954	24FEB2018	63.852054795	18 to 65 years
05MAY1954	24FEB2018	63.852054795	18 to 65 years
10NOV1966	24AUG2018	51.821917808	18 to 65 years
10NOV1966	24AUG2018	51.821917808	18 to 65 years
15JAN1945	24FEB2018	73.15890411	>65 years
15JAN1945	24FEB2018	73.15890411	>65 years
28NOV2003	28MAY2018	14.506849315	<18 years

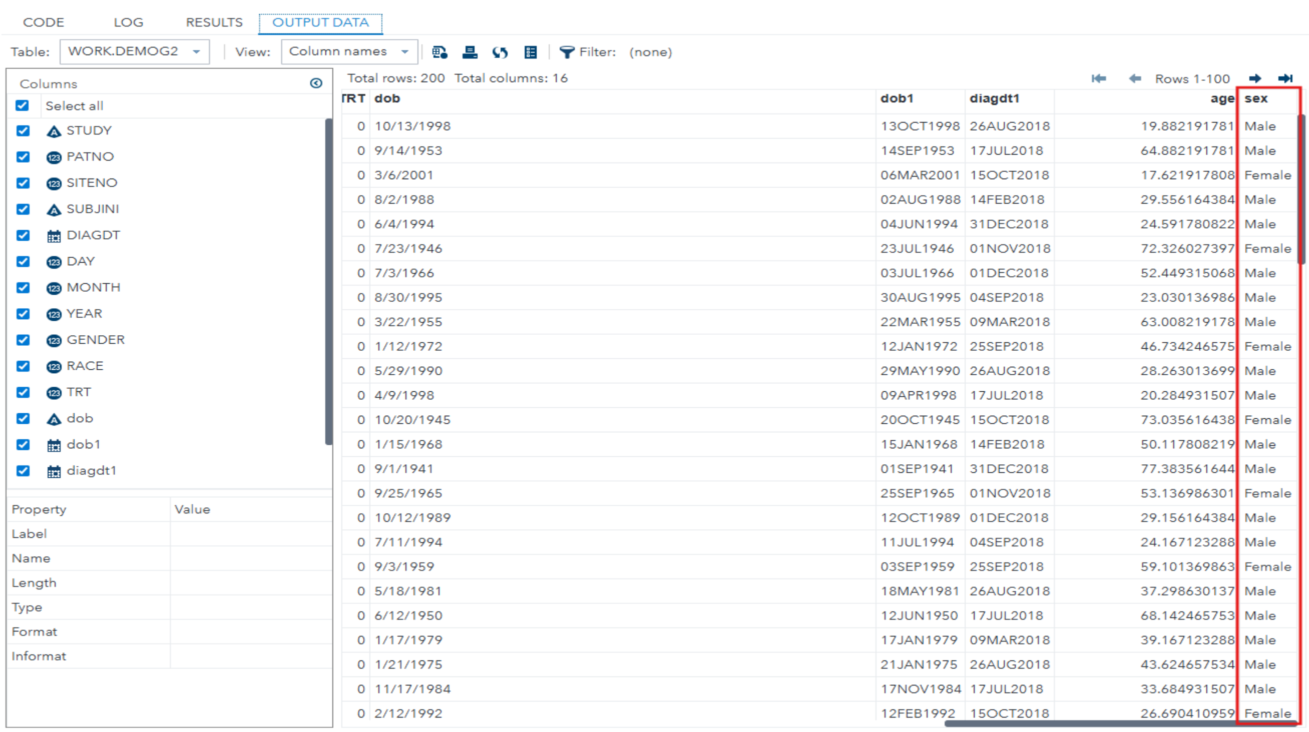
Deriving the sex variable:

The numeric variable GENDER was converted into a character variable SEX using user-defined formats. This improves readability and clarity in statistical reporting.

SAS Code:

```
proc format;

value genfmt

1='Male'

2='Female'

;

run;

data demog2;

 set demog1;

 sex=put(gender,genfmt.);

run;
```

Output:



TRT	dob	dob1	diagdt1	age	sex
0	10/13/1998	13OCT1998	26AUG2018	19.882191781	Male
0	9/14/1953	14SEP1953	17JUL2018	64.882191781	Male
0	3/6/2001	06MAR2001	15OCT2018	17.621917808	Female
0	8/2/1988	02AUG1988	14FEB2018	29.556164384	Male
0	6/4/1994	04JUN1994	31DEC2018	24.591780822	Male
0	7/23/1946	23JUL1946	01NOV2018	72.326027397	Female
0	7/3/1966	03JUL1966	01DEC2018	52.449315068	Male
0	8/30/1995	30AUG1995	04SEP2018	23.030136986	Male
0	3/22/1955	22MAR1955	09MAR2018	63.008219178	Male
0	1/12/1972	12JAN1972	25SEP2018	46.734246575	Female
0	5/29/1990	29MAY1990	26AUG2018	28.263013699	Male
0	4/9/1998	09APR1998	17JUL2018	20.284931507	Male
0	10/20/1945	20OCT1945	15OCT2018	73.035616438	Female
0	1/15/1968	15JAN1968	14FEB2018	50.117808219	Male
0	9/1/1941	01SEP1941	31DEC2018	77.383561644	Male
0	9/25/1965	25SEP1965	01NOV2018	53.136986301	Female
0	10/12/1989	12OCT1989	01DEC2018	29.156164384	Male
0	7/11/1994	11JUL1994	04SEP2018	24.167123288	Male
0	9/3/1959	03SEP1959	25SEP2018	59.101369863	Female
0	5/18/1981	18MAY1981	26AUG2018	37.298630137	Male
0	6/12/1950	12JUN1950	17JUL2018	68.142465753	Male
0	1/17/1979	17JAN1979	09MAR2018	39.167123288	Male
0	1/21/1975	21JAN1975	26AUG2018	43.624657534	Male
0	11/17/1984	17NOV1984	17JUL2018	33.684931507	Male
0	2/12/1992	12FEB1992	15OCT2018	26.690410959	Female

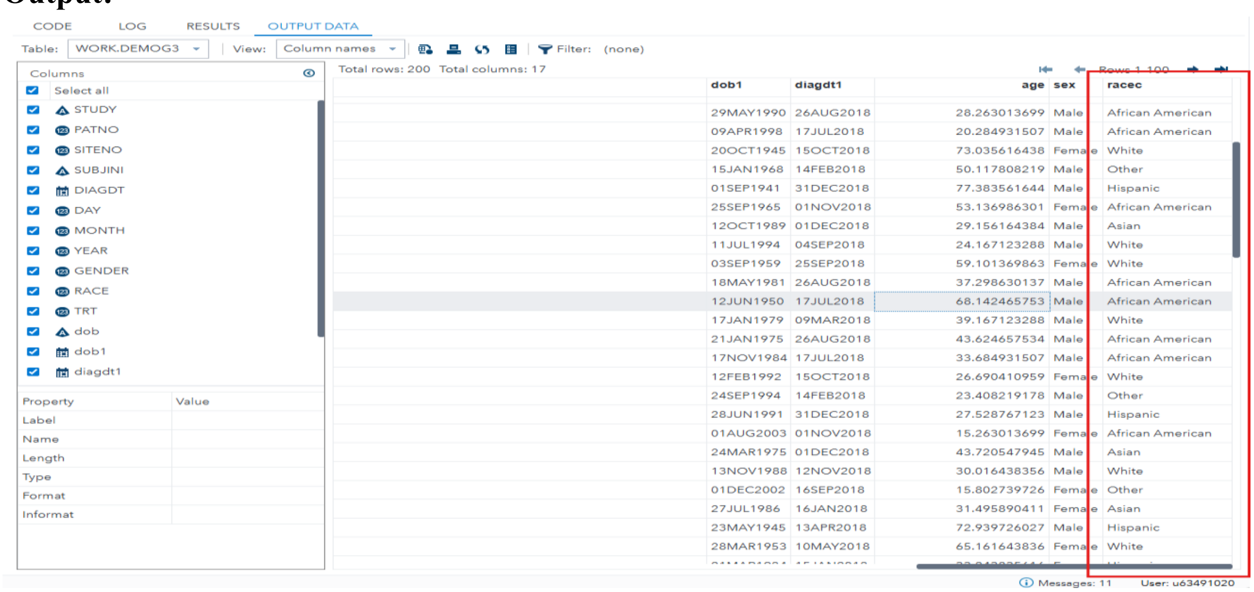
Deriving the racec variable:

The numeric variable RACE was converted into a character variable RACEC using user-defined formats to provide meaningful category labels.

SAS Code:

```
proc format;

value racefmt

1='White'

2='African American'

3='Hispanic'

4='Asian'

5='Other'

;

run;

data demog3;

 set demog2;

 racec=put(race,racefmt.);

run;
```

Output:



dob1	diagdt1	age	sex	racec
29MAY1990	26AUG2018	28.263013699	Male	African American
09APR1998	17JUL2018	20.284931507	Male	African American
20OCT1945	15OCT2018	73.035616438	Female	White
15JAN1968	14FEB2018	50.117808219	Male	Other
01SEP1941	31DEC2018	77.383561644	Male	Hispanic
25SEP1965	01NOV2018	53.136986301	Female	African American
12OCT1989	01DEC2018	29.156164384	Male	Asian
11JUL1994	04SEP2018	24.167123288	Male	White
03SEP1959	25SEP2018	59.101369863	Female	White
18MAY1981	26AUG2018	37.298630137	Male	African American
12JUN1950	17JUL2018	68.142465753	Male	African American
17JAN1979	09MAR2018	39.167123288	Male	White
21JAN1975	26AUG2018	43.624657534	Male	African American
17NOV1984	17JUL2018	33.684931507	Male	African American
12FEB1992	15OCT2018	26.690410959	Female	White
24SEP1994	14FEB2018	23.408219178	Male	Other
28JUN1991	31DEC2018	27.528767123	Male	Hispanic
01AUG2003	01NOV2018	15.263013699	Female	African American
24MAR1975	01DEC2018	43.720547945	Male	Asian
13NOV1988	12NOV2018	30.016438356	Male	White
01DEC2002	16SEP2018	15.802739726	Female	Other
27JUL1986	16JAN2018	31.495890411	Female	Asian
23MAY1945	13APR2018	72.939726027	Male	Hispanic
28MAR1953	10MAY2018	65.161643836	Female	White

Interpretation:

The variables Age, Sex, and Racec were derived to facilitate meaningful statistical analysis and reporting.

- Age was calculated using the difference between Diagnosis Date and Date of Birth, providing a continuous variable for demographic comparison.
- Sex was derived by converting numeric gender codes into descriptive categories (Male/Female) using a user-defined format.
- Racec was created by mapping numeric race codes to labeled race categories such as White, African American, Hispanic, Asian, and Other.

These derived variables improve data interpretability and enable appropriate descriptive and inferential statistical analysis in clinical trial reporting.

5.4 Construction of Treatment Groups

To generate overall summary statistics, the dataset was duplicated and assigned a new treatment value (TRT = 2). The datasets were combined to allow reporting by:

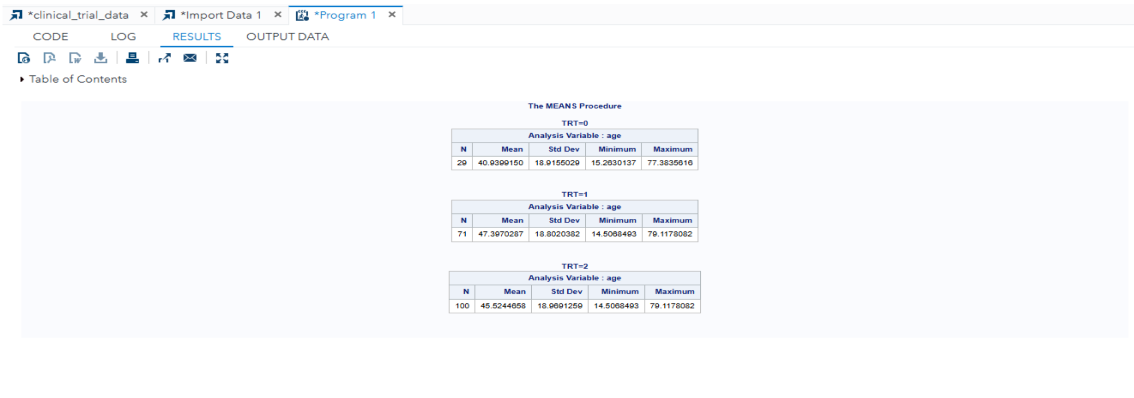
- Placebo
- Active Treatment
- Total Population

SAS code:

```
data demog1;
 set demog;
 dob=compress(cat(month,'/',day,'/',year));
 dob1=input(dob,mmddyy10.);
 format dob1 date9.;
 diagdt1 = diagdt;
 format diagdt1 date9.;
 age=(diagdt1-dob1)/365;
 output;
 trt=2;
 output;
```

run;

Output:


*clinical_trial_data *Import Data 1 *Program 1

CODE LOG RESULTS OUTPUT DATA

Table of Contents

The MEANS Procedure

TRT=0

Analysis Variable : age				
N	Mean	Std Dev	Minimum	Maximum
29	40.9399150	18.9155029	15.2630137	77.3835616

TRT=1

Analysis Variable : age				
N	Mean	Std Dev	Minimum	Maximum
71	47.3970287	18.8020382	14.5068493	79.1178082

TRT=2

Analysis Variable : age				
N	Mean	Std Dev	Minimum	Maximum
100	45.5244658	18.9691259	14.5068493	79.1178082

Interpretation:

In this step, the dataset was modified to create a Total Population group for reporting purposes.

First, the age variable was derived using Date of Birth and Diagnosis Date. Then, the OUTPUT statement was used to duplicate each observation.

For the duplicated observation, the treatment variable (TRT) was reassigned a value of 2, representing the Total Population group.

As a result, each subject appears twice in the dataset:

- Once under their original treatment group (Placebo or Active Treatment)
- Once under the Total Population group (TRT = 2)

This approach enables the generation of summary statistics separately for:

- Placebo group
- Active Treatment group
- Overall Total Population

This structure is commonly used in clinical trial reporting to produce combined summary tables.

5.5 Creation of Summary Datasets

Obtaining the summary stats for age:

The dataset was first sorted by treatment group (TRT). Then, PROC MEANS was used to compute summary statistics for the continuous variable AGE separately for each treatment group. The results were stored in a new dataset named AGESTATS.

SAS Code:

```
proc sort data=demog1;

by trt;

run;

proc means data=demog1;

var age;

output out=agestats;

by trt;

run;
```

Output:

The MEANS Procedure

TRT=0

Analysis Variable : age				
N	Mean	Std Dev	Minimum	Maximum
29	40.9399150	18.9155029	15.2630137	77.3835616

TRT=1

Analysis Variable : age				
N	Mean	Std Dev	Minimum	Maximum
71	47.3970287	18.8020382	14.5068493	79.1178082

TRT=2

Analysis Variable : age				
N	Mean	Std Dev	Minimum	Maximum
100	45.5244658	18.9691259	14.5068493	79.1178082

Obtaining the summary stats for ageGroup:

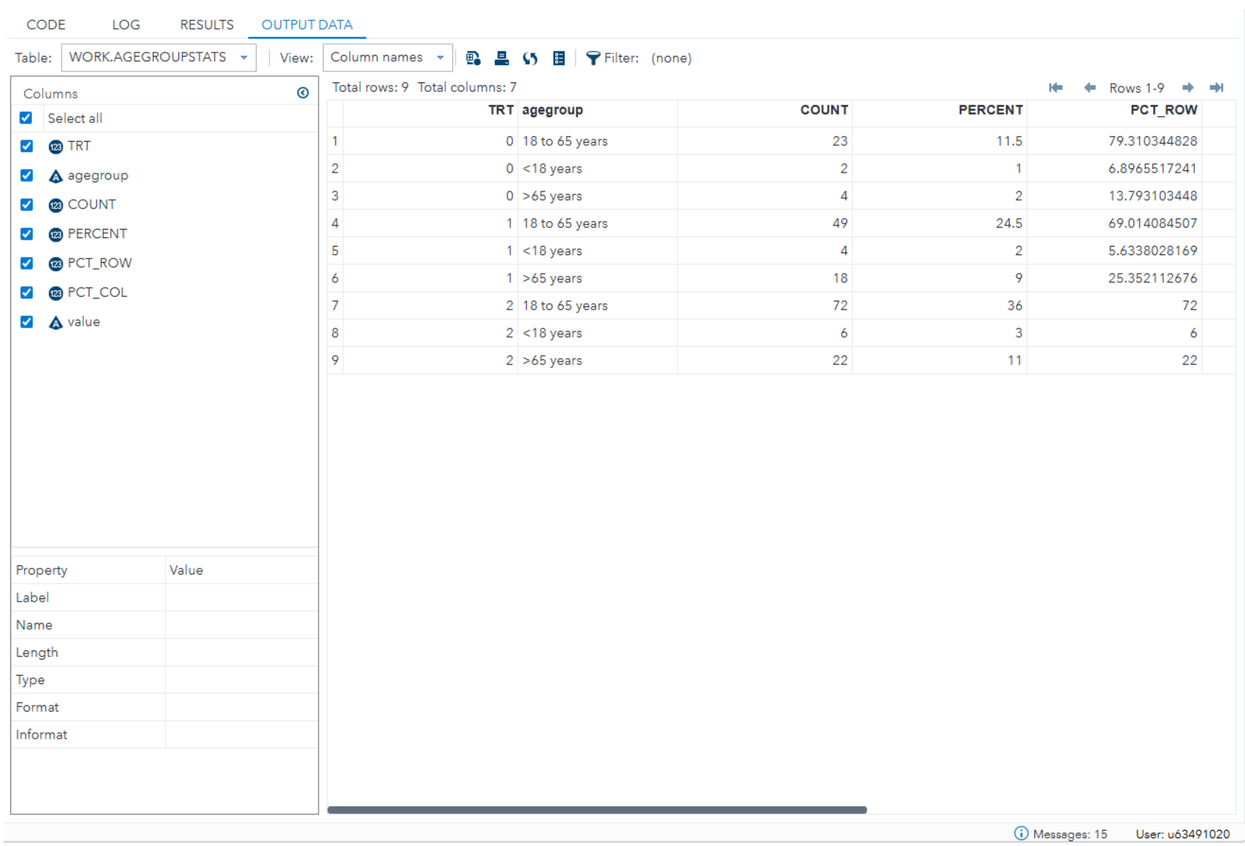
Description:

PROC FREQ was used to generate the distribution of age groups by treatment, and the OUTPCT option calculated row percentages. The output dataset agegroupstats was then formatted to display results as count and percentage in the form *n (%)* for reporting in the demographic table.

SAS code:

```
proc freq data=demog4 noprint;
   tables trt*agegroup / outpct out=agegroupstats;
run;
data agegroupstats;
   set agegroupstats;
   value = cat(count,' (',strip(put(round(pct_row,.1),8.1)),'%)');
run;
```

Output:



	TRT	agegroup	COUNT	PERCENT	PCT_ROW
1	0	18 to 65 years	23	11.5	79.310344828
2	0	<18 years	2	1	6.8965517241
3	0	>65 years	4	2	13.793103448
4	1	18 to 65 years	49	24.5	69.014084507
5	1	<18 years	4	2	5.6338028169
6	1	>65 years	18	9	25.352112676
7	2	18 to 65 years	72	36	72
8	2	<18 years	6	3	6
9	2	>65 years	22	11	22

Obtaining the summary stats for sex:

Description:

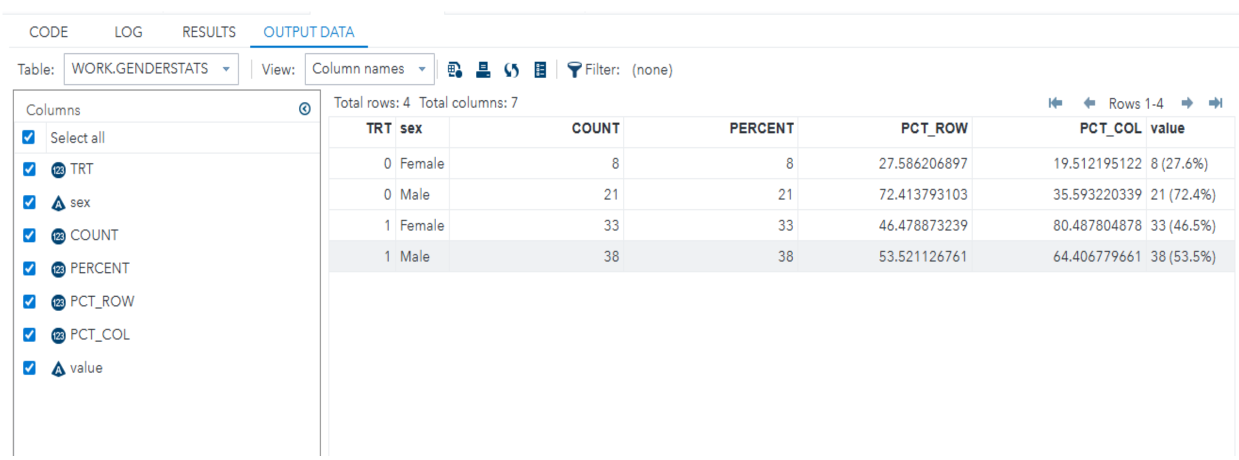
PROC FREQ was used to calculate the frequency distribution of SEX across treatment groups. The NOPRINT option suppressed printed output and stored results in the dataset GENDERSTATS.

The count and row percentage were then combined into a single formatted variable (VALUE) for reporting purposes.

SAS code:

```
proc freq data=demog2 noprint;

table trt*sex/outpct out=genderstats;

run;

/*Concatenating the COUNT and PERCENT variables*/

data genderstats;

 set genderstats;

 value=cat(count,' (',round(pct_row,.1),'%)');

run;
```

Output:



TRT	sex	COUNT	PERCENT	PCT_ROW	PCT_COL	value
0	Female	8	8	27.586206897	19.512195122	8 (27.6%)
0	Male	21	21	72.413793103	35.593220339	21 (72.4%)
1	Female	33	33	46.478873239	80.487804878	33 (46.5%)
1	Male	38	38	53.521126761	64.406779661	38 (53.5%)

Obtaining the summary stats for recec:

Description:

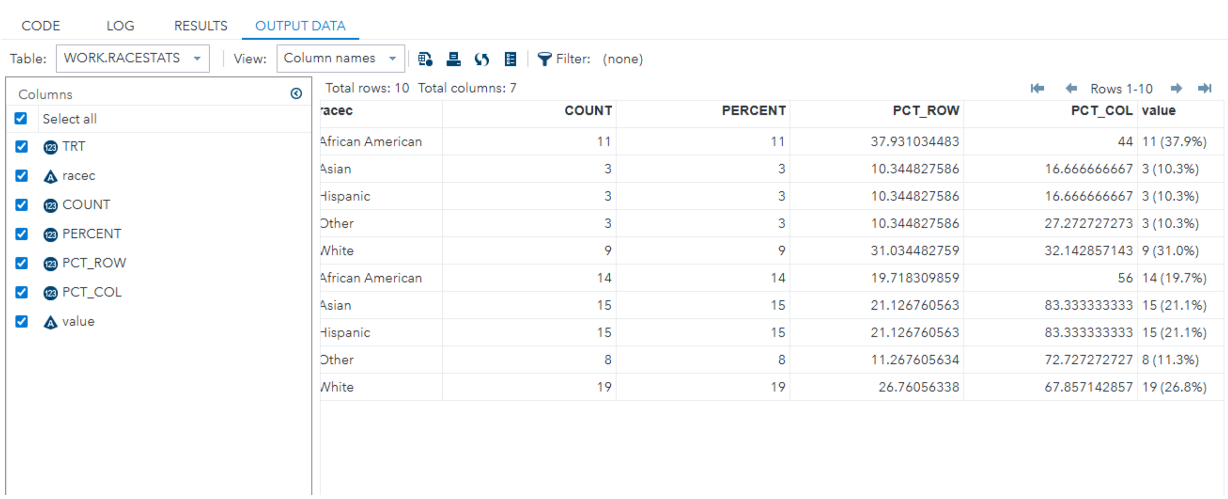
PROC FREQ was used to compute frequency distribution of RACEC across treatment groups. The results were stored in RACESTATS.

The count and row percentage were concatenated into a formatted reporting variable.

SAS code:

```
proc freq data=demog3 noprint;

table trt*racec/outpct out=racestats;

run;


data racestats;

 set racestats;

 value=cat(count,' (',strip(put(round(pct_row,.1),8.1)),'%)');

run;
```

Output:



CODE LOG RESULTS OUTPUT DATA

Table: WORK.RACESTATS View: Column names Filter: (none)

Columns: Select all, TRT, racec, COUNT, PERCENT, PCT_ROW, PCT_COL, value

Total rows: 10 Total columns: 7 Rows 1-10

racec	COUNT	PERCENT	PCT_ROW	PCT_COL	value
African American	11	11	37.931034483	44	11 (37.9%)
Asian	3	3	10.344827586	16.666666667	3 (10.3%)
Hispanic	3	3	10.344827586	16.666666667	3 (10.3%)
Other	3	3	10.344827586	27.272727273	3 (10.3%)
White	9	9	31.034482759	32.142857143	9 (31.0%)
African American	14	14	19.718309859	56	14 (19.7%)
Asian	15	15	21.126760563	83.333333333	15 (21.1%)
Hispanic	15	15	21.126760563	83.333333333	15 (21.1%)
Other	8	8	11.267605634	72.727272727	8 (11.3%)
White	19	19	26.76056338	67.857142857	19 (26.8%)

5.6 Stacking all summary statistics together:

Stacking all three summary statistics together means combining the separate outputs for:

- Age (continuous variable summary)
- Gender (categorical frequency summary)
- Race (categorical frequency summary)

into one consolidated table.

5.6.1 Renaming the variables:

In the data preparation step, variables were renamed to improve clarity and to make them suitable for reporting in the final demographic table.

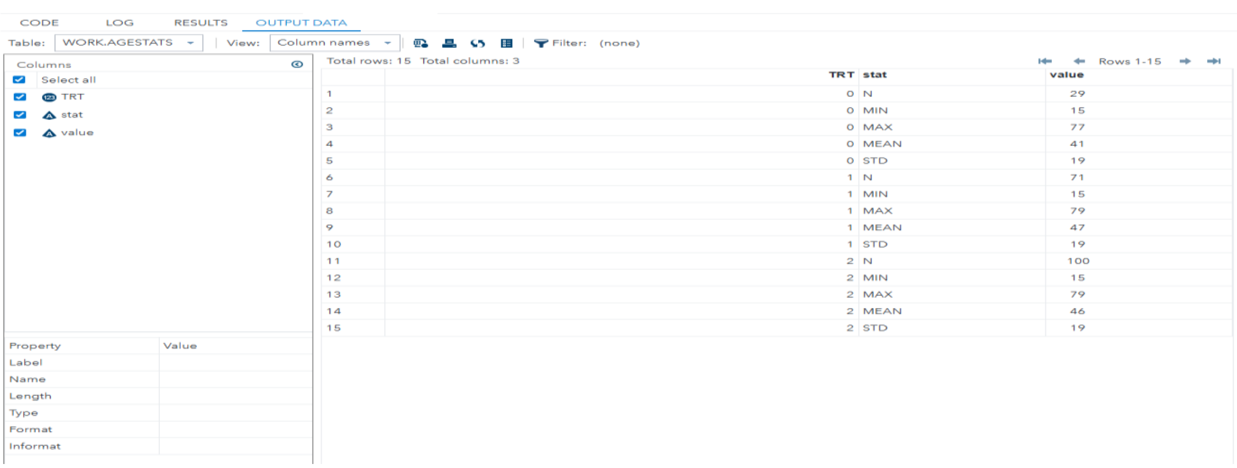
In our data we are renaming the fallowing variables

a.In agestats Renaming _stat_ to stat

SAS code:

```
data agestats;
 set agestats;
 value=put(age,8.);
 rename _stat_=stat;
 drop _type_ _freq_ age;
run;
```

Output:



	TRT	stat	value
1	0	N	29
2	0	MIN	15
3	0	MAX	77
4	0	MEAN	41
5	0	STD	19
6	1	N	71
7	1	MIN	15
8	1	MAX	79
9	1	MEAN	47
10	1	STD	19
11	2	N	100
12	2	MIN	15
13	2	MAX	79
14	2	MEAN	46
15	2	STD	19

b. In agegroupstats renaming agegroup to stat

SAS code:

```
data agegroupstats;
    set agegroupstats;
    value = cat(count,' (',strip(put(round(pct_row,.1),8.1)),'%)');
    rename agegroup=stat;
    drop count percent pct_row pct_col;
run;
```

Output:

Table: WORK.AGEGROUPSTATS View: Column names Filter: (none)

Columns: Select all, TRT, stat, value

Total rows: 9 Total columns: 3 Rows 1-9

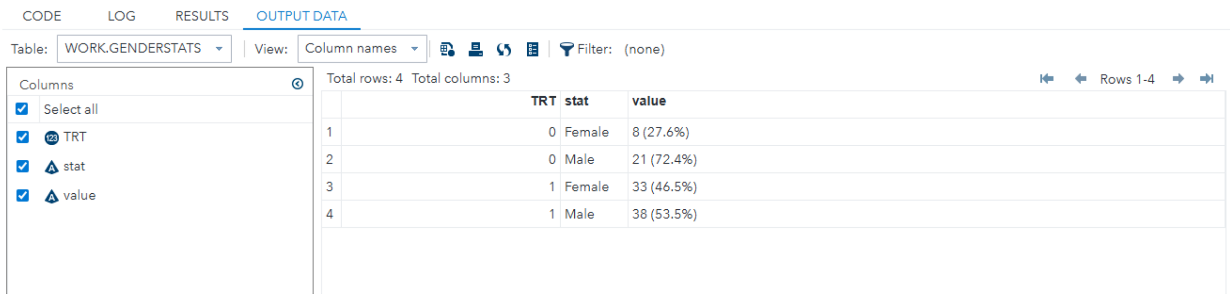
	TRT	stat	value
1	0	18 to 65 years	23 (79.3%)
2	0	<18 years	2 (6.9%)
3	0	>65 years	4 (13.8%)
4	1	18 to 65 years	49 (69.0%)
5	1	<18 years	4 (5.6%)
6	1	>65 years	18 (25.4%)
7	2	18 to 65 years	72 (72.0%)
8	2	<18 years	6 (6.0%)
9	2	>65 years	22 (22.0%)

c.In genderstats renaming sex to stat:

SAS code:

```
data genderstats;
 set genderstats;
 value=cat(count,' (',round(pct_row,.1),'%)');
 rename sex=stat;
 drop count percent pct_row pct_col;
run;
```

Output:

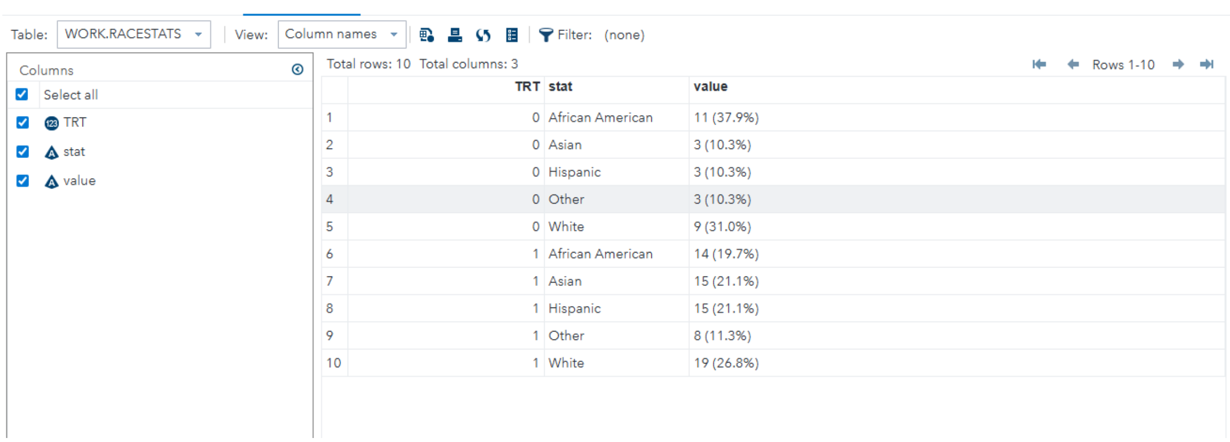


	TRT	stat	value
1	0	Female	8 (27.6%)
2	0	Male	21 (72.4%)
3	1	Female	33 (46.5%)
4	1	Male	38 (53.5%)

d.In racestats renaming racec to stat:

```
data racestats;
 set racestats;
 value=cat(count,' (',strip(put(round(pct_row,.1),8.1)),'%)');
 rename racec=stat;
 drop count percent pct_row pct_col;
run;
```

Output:



	TRT	stat	value
1	0	African American	11 (37.9%)
2	0	Asian	3 (10.3%)
3	0	Hispanic	3 (10.3%)
4	0	Other	3 (10.3%)
5	0	White	9 (31.0%)
6	1	African American	14 (19.7%)
7	1	Asian	15 (21.1%)
8	1	Hispanic	15 (21.1%)
9	1	Other	8 (11.3%)
10	1	White	19 (26.8%)

Interpretation:

In the data preparation step, the variables _STAT_, agegroup, sex, and racec were renamed to a common variable name stat.

This was done to standardize the structure of all summary datasets before combining them into a single demographic table.

Each procedure (PROC MEANS and PROC FREQ) generates different variable names for row labels. If these variable names are different, the datasets cannot be properly stacked using the SET statement.

By renaming:

- _STAT_ (from agestats) to stat
- agegroup to stat
- sex to stat
- racec to stat

all datasets were made consistent.

5.6.2 Appending all stats together:

In this step, all individual summary statistic datasets were combined into a single dataset named allstats.

The datasets appended were:

- agestats (age summary statistics)
- agegroupstats (age group frequencies)
- genderstats (gender frequencies)
- racestats (race frequencies)

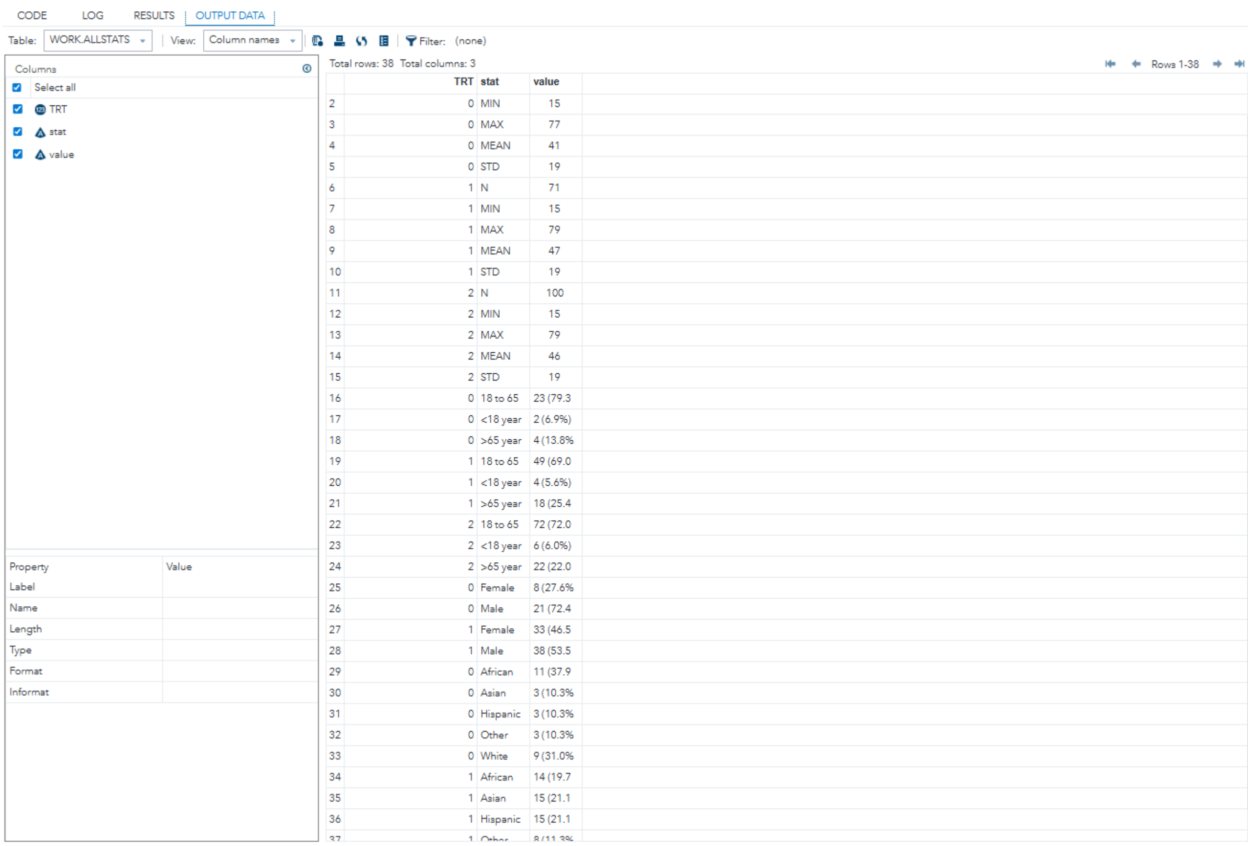
The SET statement vertically stacks these datasets one below another.

This process is known as appending or stacking datasets.

SAS code:

```
data allstats;
 set agestats agegroupstats genderstats racestats ;
run;
```

Output:



	TRT	stat	value
2	0	MIN	15
3	0	MAX	77
4	0	MEAN	41
5	0	STD	19
6	1	N	71
7	1	MIN	15
8	1	MAX	79
9	1	MEAN	47
10	1	STD	19
11	2	N	100
12	2	MIN	15
13	2	MAX	79
14	2	MEAN	46
15	2	STD	19
16	0	18 to 65	23 (79.3
17	0	<18 year	2 (6.9%)
18	0	>65 year	4 (13.8%
19	1	18 to 65	49 (69.0
20	1	<18 year	4 (5.6%)
21	1	>65 year	18 (25.4
22	2	18 to 65	72 (72.0
23	2	<18 year	6 (6.0%)
24	2	>65 year	22 (22.0
25	0	Female	8 (27.6%
26	0	Male	21 (72.4
27	1	Female	33 (46.5
28	1	Male	38 (53.5
29	0	African	11 (37.9
30	0	Asian	3 (10.3%
31	0	Hispanic	3 (10.3%
32	0	Other	3 (10.3%
33	0	White	9 (31.0%
34	1	African	14 (19.7
35	1	Asian	15 (21.1
36	1	Hispanic	15 (21.1
37	1	Other	8 (11.3%

Interpretation

Appending all statistics together created one consolidated dataset containing:

- Continuous variable summaries (Age)
- Categorical variable summaries (Age Group, Gender, Race)
- Corresponding treatment group values
- Ordering variables (ord, subord)

5.6.3 Fixing the precision points:

In this step, the numeric age summary statistics were formatted to control the number of decimal places displayed in the final demographic table.

The variable age was converted into a character variable named value using the PUT() function, and different precision levels were applied depending on the statistic:

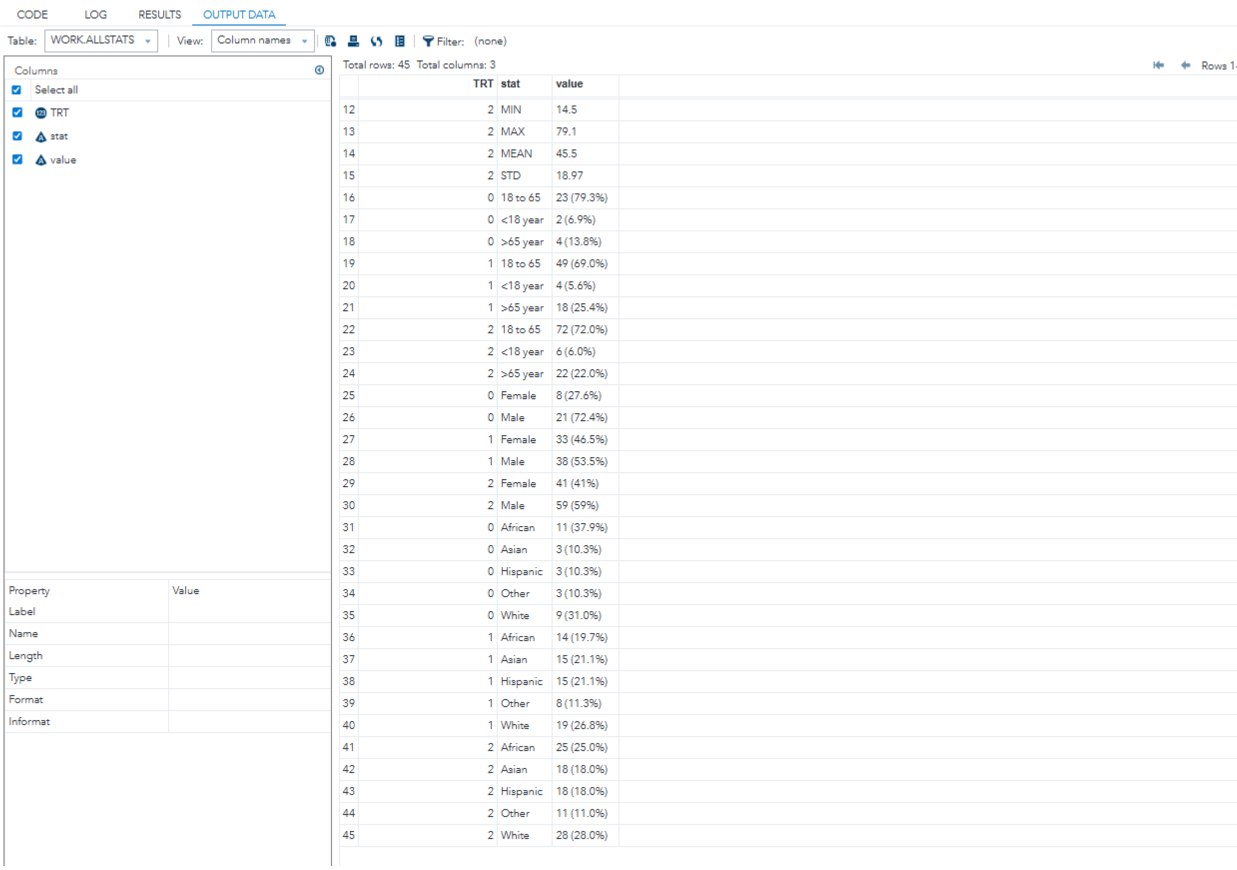
- N → No decimal places
- MEAN → 1 decimal place
- STD → 2 decimal places
- MIN → 1 decimal place
- MAX → 1 decimal place

The STRIP() function was used to remove leading and trailing spaces.

SAS Code:

```
data agestats;
   set agestats;
   length value $10.;
   if _stat_ = 'N' then value = strip(put(age, 8.));
   else if _stat_ = 'MEAN' then value = strip(put(age, 8.1));
   else if _stat_ = 'STD' then value = strip(put(age,8.2));
   else if _stat_ = 'MIN' then value = strip(put(age, 8.1));
   else if _stat_ = 'MAX' then value = strip(put(age, 8.1));

   rename _stat_ = stat;
   drop _type_ _freq_ age;
run;
```

Output:



CODE LOG RESULTS OUTPUT DATA

Table: WORK.ALLSTATS View: Column names Filter: (none)

Columns: Select all, TRT, stat, value

Property | Value: Label, Name, Length, Type, Format, Informat

Total rows: 45 Total columns: 3 Rows 1-4

	TRT	stat	value
12	2	MIN	14.5
13	2	MAX	79.1
14	2	MEAN	45.5
15	2	STD	18.97
16	0	18 to 65	23 (79.3%)
17	0	<18 year	2 (6.9%)
18	0	>65 year	4 (13.8%)
19	1	18 to 65	49 (69.0%)
20	1	<18 year	4 (5.6%)
21	1	>65 year	18 (25.4%)
22	2	18 to 65	72 (72.0%)
23	2	<18 year	6 (6.0%)
24	2	>65 year	22 (22.0%)
25	0	Female	8 (27.6%)
26	0	Male	21 (72.4%)
27	1	Female	33 (46.5%)
28	1	Male	38 (53.5%)
29	2	Female	41 (41%)
30	2	Male	59 (59%)
31	0	African	11 (37.9%)
32	0	Asian	3 (10.3%)
33	0	Hispanic	3 (10.3%)
34	0	Other	3 (10.3%)
35	0	White	9 (31.0%)
36	1	African	14 (19.7%)
37	1	Asian	15 (21.1%)
38	1	Hispanic	15 (21.1%)
39	1	Other	8 (11.3%)
40	1	White	19 (26.8%)
41	2	African	25 (25.0%)
42	2	Asian	18 (18.0%)
43	2	Hispanic	18 (18.0%)
44	2	Other	11 (11.0%)
45	2	White	28 (28.0%)

Interpretation

This step ensures consistent and professional presentation of numerical results.

It prevents:

• Excessive decimal places
• Inconsistent formatting
• Unnecessary numeric clutter in the final table

5.6.4 Assigning Ordering Variables (ord and subord)

In this step, the variables ord and subord were created to control the display order of statistics in the final demographic table.

The variable ord represents the main category order (for example, Age section = 1), while subord represents the internal ordering of statistics within that category.

a.agestats

For the age statistics:

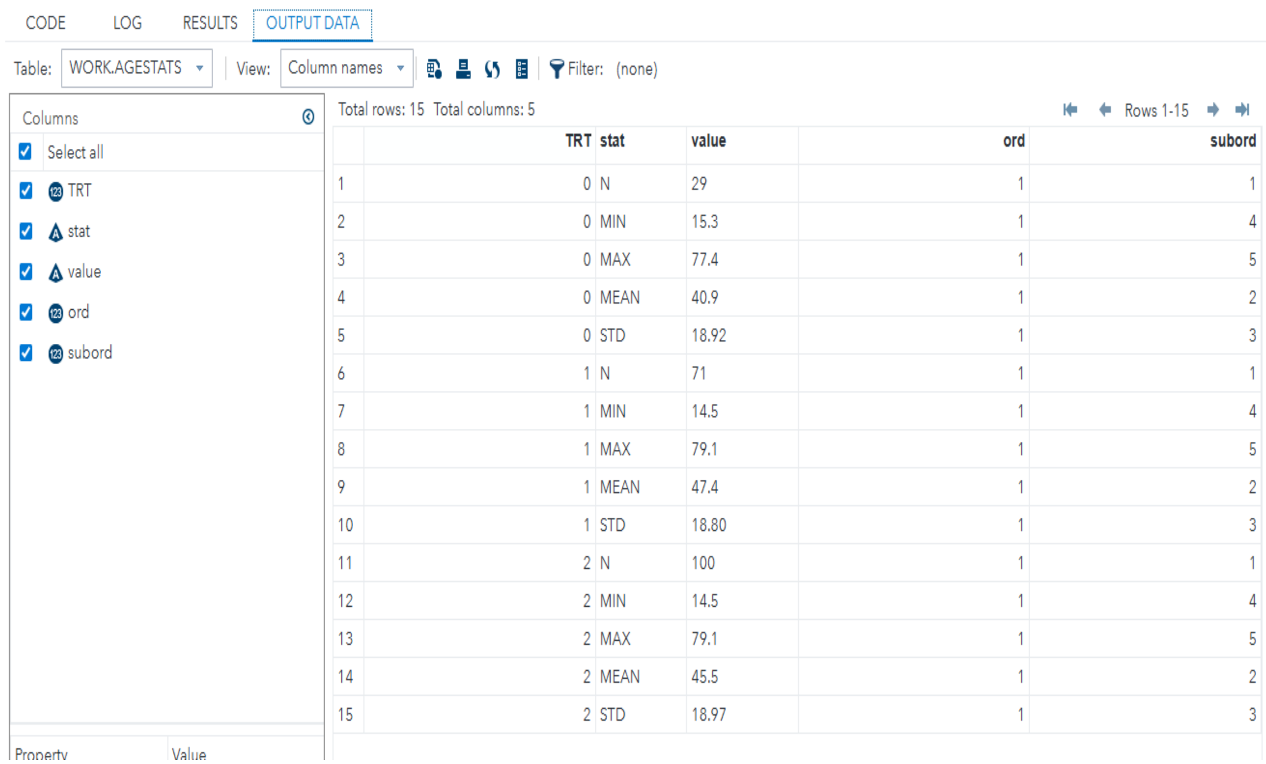
- ord = 1 was assigned to indicate the Age section.
- subord was assigned as:
  - 1 → N
  - 2 → Mean
  - 3 → Standard Deviation
  - 4 → Minimum
  - 5 → Maximum

SAS Code:

```
proc means data=demog1;
var age;
output out=agestats;
by trt;
run;
data agestats;
   set agestats;
   length value $10.;
   ord=1;
   if _stat_ = 'N' then do; subord=1; value = strip(put(age, 8.)); end;
   else if _stat_='MEAN' then do; subord=2; value = strip(put(age, 8.1)); end;
   else if _stat_='STD' then do; subord=3; value = strip(put(age,8.2)); end;
   else if _stat_='MIN' then do; subord=4; value = strip(put(age, 8.1)); end;
```

```
    else if _stat_='MAX' then do; subord = 5; value = strip(put(age, 8.1));end;
    rename _stat_ = stat;
    drop _type_ _freq_ age;
run;
```

Out put:



	TRT	stat	value	ord	subord
1	0	N	29	1	1
2	0	MIN	15.3	1	4
3	0	MAX	77.4	1	5
4	0	MEAN	40.9	1	2
5	0	STD	18.92	1	3
6	1	N	71	1	1
7	1	MIN	14.5	1	4
8	1	MAX	79.1	1	5
9	1	MEAN	47.4	1	2
10	1	STD	18.80	1	3
11	2	N	100	1	1
12	2	MIN	14.5	1	4
13	2	MAX	79.1	1	5
14	2	MEAN	45.5	1	2
15	2	STD	18.97	1	3

b. agegroupstats

The variable ord = 2 was assigned to indicate that Age Group belongs to the second section of the demographic table.

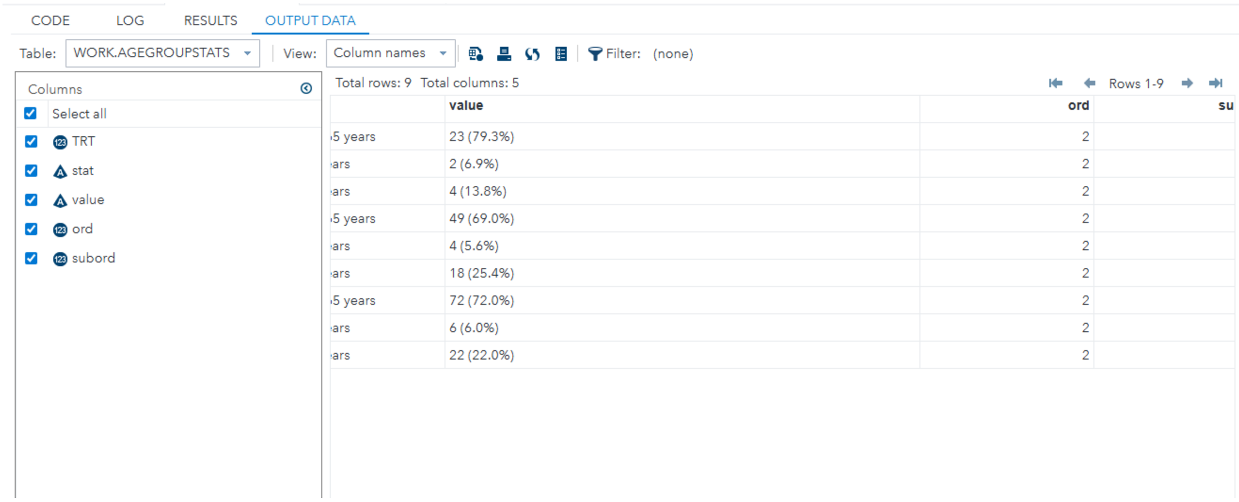
The variable subord was assigned to control the order of age categories:

- 1 → <18 years
- 2 → 18 to 65 years
- 3 → >65 years

SAS Code:

```
proc freq data=demog4 noprint;
   tables trt*agegroup / outpct out=agegroupstats;
run;
data agegroupstats;
   set agegroupstats;
   value = cat(count,' (',strip(put(round(pct_row,.1),8.1)),'%)');
   /* Order for demographic table */
   ord = 2;
   if agegroup = '<18 years' then subord = 1;
   else if agegroup = '18 to 65 years' then subord = 2;
   else if agegroup = '>65 years' then subord = 3;
   rename agegroup = stat;
   drop count percent pct_row pct_col;
run;
```

Output:



CODE LOG RESULTS OUTPUT DATA

Table: WORK.AGEGROUPSTATS View: Column names Filter: (none)

Columns: Select all, TRT, stat, value, ord, subord

Total rows: 9 Total columns: 5 Rows 1-9

	value	ord	su
5 years	23 (79.3%)	2	
ars	2 (6.9%)	2	
ars	4 (13.8%)	2	
5 years	49 (69.0%)	2	
ars	4 (5.6%)	2	
ars	18 (25.4%)	2	
5 years	72 (72.0%)	2	
ars	6 (6.0%)	2	
ars	22 (22.0%)	2	

c. genderstats

The variable ord was assigned to indicate the main section placement of Gender within the demographic table.

The variable subord was assigned to control the internal ordering of gender categories:

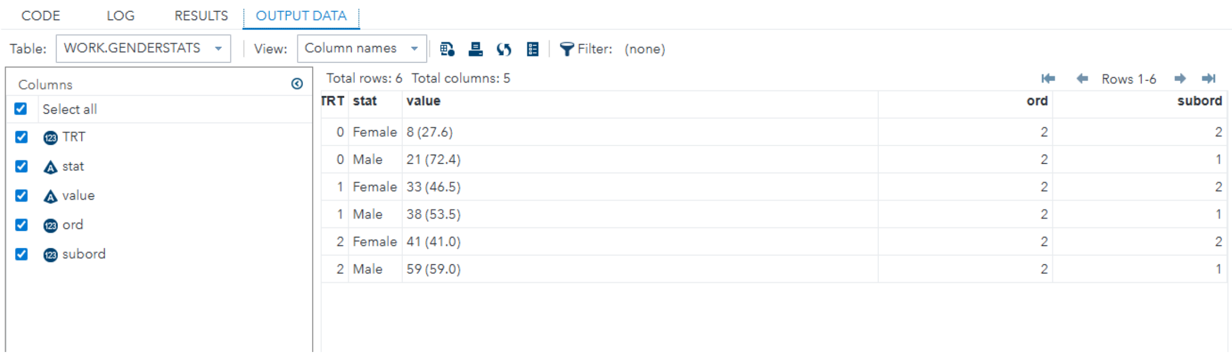
- 1 → Male
- 2 → Female

SAS Code:

```
proc freq data=demog2 noprint;
table trt*sex/outpct out=genderstats;
run;

data genderstats;
   set genderstats;
   value = cat(count,' (', strip(put(round(pct_row,.1),8.1)),')');
   ord = 3;
```

```
    if sex='Male' then subord=1;
    else subord=2;
    rename sex=stat;
    drop count percent pct_row pct_col;
run;
```

Output:



TRT	stat	value	ord	subord
0	Female	8 (27.6)	2	2
0	Male	21 (72.4)	2	1
1	Female	33 (46.5)	2	2
1	Male	38 (53.5)	2	1
2	Female	41 (41.0)	2	2
2	Male	59 (59.0)	2	1

d. racestats

The variable ord = 4 was assigned to indicate that Race belongs to the fourth section of the demographic table.

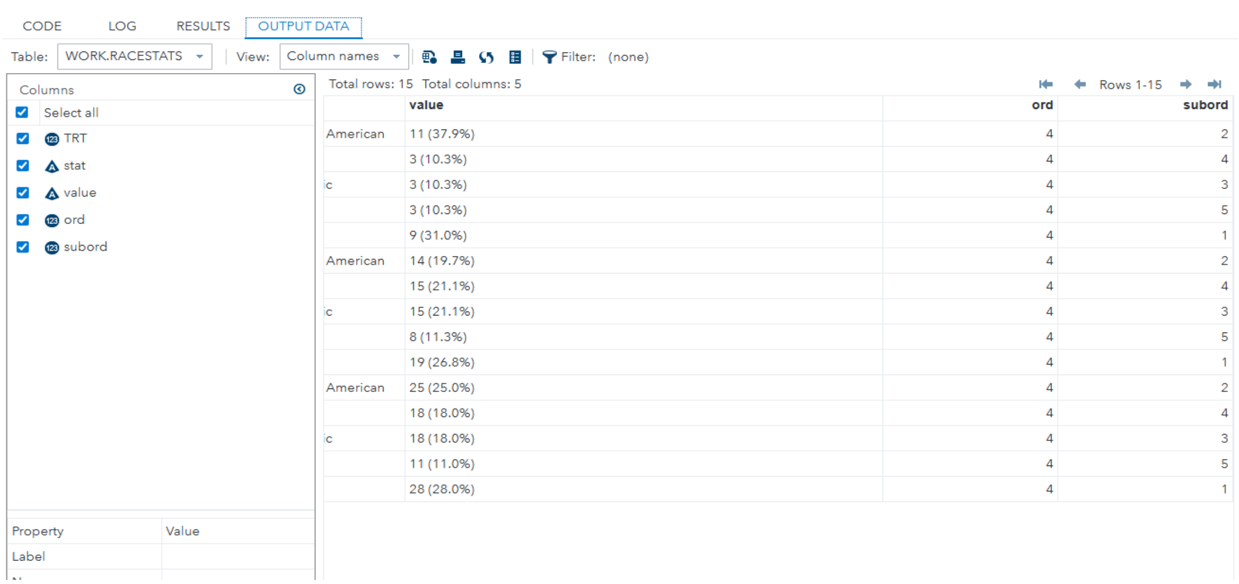
The variable subord was assigned to control the display order of race categories as follows:

- 1 → White
- 2 → African American
- 3 → Hispanic
- 4 → Asian
- 5 → Other

SAS code:

```
proc freq data=demog3 noprint;
table trt*racec/outpct out=racestats;
run;

data racestats;
 set racestats;
 value=cat(count,' (',strip(put(round(pct_row,.1),8.1)),'%)');
 ord=4;
 if racec='White' then subord=1;
 else if racec='African American' then subord=2;
 else if racec='Hispanic' then subord=3;
 else if racec='Asian' then subord=4;
 else if racec='Other' then subord=5;
 rename racec=stat;
 drop count percent pct_row pct_col;
run;
```

Output:



CODE LOG RESULTS OUTPUT DATA

Table: WORK.RACESTATS View: Column names Filter: (none)

Columns
Select all
TRT
stat
value
ord
subord

Property Value
Label

Total rows: 15 Total columns: 5 Rows 1-15

	value	ord	subord
American	11 (37.9%)	4	2
	3 (10.3%)	4	4
c	3 (10.3%)	4	3
	3 (10.3%)	4	5
	9 (31.0%)	4	1
American	14 (19.7%)	4	2
	15 (21.1%)	4	4
c	15 (21.1%)	4	3
	8 (11.3%)	4	5
	19 (26.8%)	4	1
American	25 (25.0%)	4	2
	18 (18.0%)	4	4
c	18 (18.0%)	4	3
	11 (11.0%)	4	5
	28 (28.0%)	4	1

5.6.5 Transposing the dataset

In this step, PROC TRANSPOSE was used to restructure the combined dataset allstats from long format to wide format.

Before transposition:

- Each row contained:
    - ord
    - subord
    - stat
    - trt
    - value

So treatment groups (0, 1, 2) were stored as rows.

Using PROC TRANSPOSE:

- The variable value was transposed.
- The id trt; statement converted treatment group values (0, 1, 2) into separate columns.
- The by ord subord stat; statement ensured that rows remained grouped and ordered properly.
- The option prefix=_ created new column names as _0, _1, and _2.

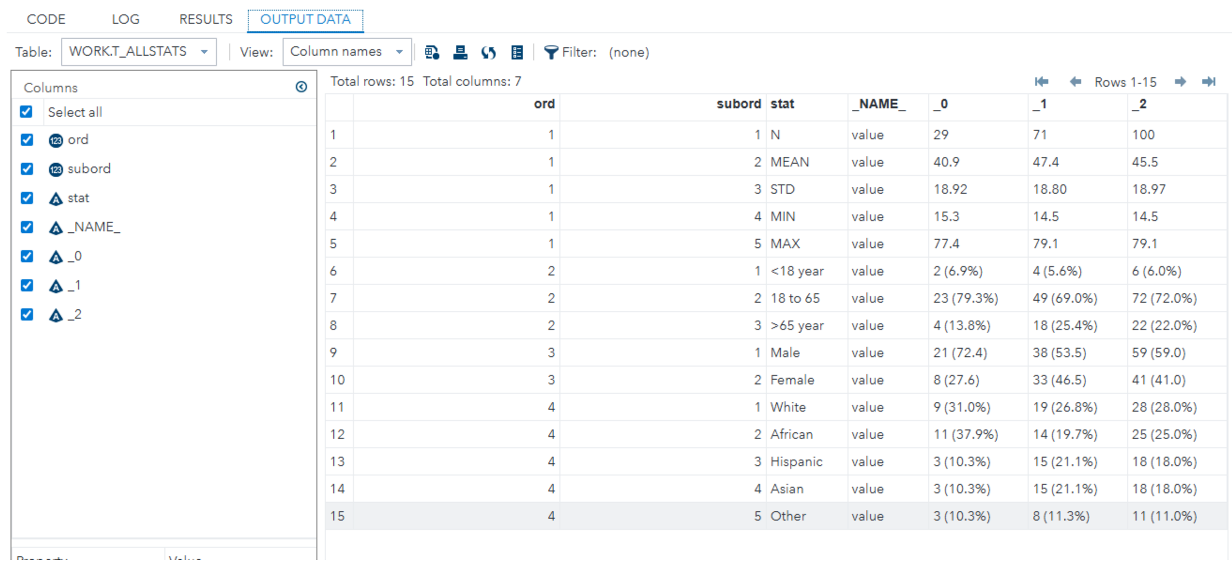
SAS Code:

```
data allstats;
 set agestats agegroupstats genderstats racestats ;
run;
proc sort data=allstats;
by ord subord stat;
run;
proc transpose data = allstats out = t_allstats prefix=_;
var value;
id trt;
```

by ord subord stat;

run;

Out put:



	ord	subord	stat	_NAME_	_0	_1	_2
1	1	1	N	value	29	71	100
2	1	2	MEAN	value	40.9	47.4	45.5
3	1	3	STD	value	18.92	18.80	18.97
4	1	4	MIN	value	15.3	14.5	14.5
5	1	5	MAX	value	77.4	79.1	79.1
6	2	1	<18 year	value	2 (6.9%)	4 (5.6%)	6 (6.0%)
7	2	2	18 to 65	value	23 (79.3%)	49 (69.0%)	72 (72.0%)
8	2	3	>65 year	value	4 (13.8%)	18 (25.4%)	22 (22.0%)
9	3	1	Male	value	21 (72.4)	38 (53.5)	59 (59.0)
10	3	2	Female	value	8 (27.6)	33 (46.5)	41 (41.0)
11	4	1	White	value	9 (31.0%)	19 (26.8%)	28 (28.0%)
12	4	2	African	value	11 (37.9%)	14 (19.7%)	25 (25.0%)
13	4	3	Hispanic	value	3 (10.3%)	15 (21.1%)	18 (18.0%)
14	4	4	Asian	value	3 (10.3%)	15 (21.1%)	18 (18.0%)
15	4	5	Other	value	3 (10.3%)	8 (11.3%)	11 (11.0%)

Interpretation

After transposition, the dataset structure changed from vertical to horizontal.

Now the dataset contains:

- ord
- subord
- stat
- _0 → TRT 0 (Placebo)
- _1 → TRT 1 (Active Treatment)
- _2 → TRT 2 (Total Population)

5.7 Constructing the final report:

In this step, the final demographic table was generated using PROC SQL and PROC REPORT.

First, PROC SQL was used to calculate the total number of subjects in each treatment group:

- TRT = 0 → Placebo group
- TRT = 1 → Active Treatment group
- TRT = 2 → Total population

The counts were stored in macro variables:

- &placebo
- &active
- &total

These macro variables were later dynamically inserted into the column headers of the final report to display group sample sizes.

Next, PROC REPORT was used to generate the final formatted demographic table (Table 1.1).

The dataset t_allstats (transposed dataset) was used as input.

The following structure was defined:

- ord and subord were used for row ordering (not printed).
- stat displayed the demographic characteristic name.
- _0, _1, and _2 represented:
    - Placebo (TRT 0)
    - Active Treatment (TRT 1)
    - All Patients (TRT 2)

The split='|' option allowed multi-line column headers to display:

Placebo (N=…)
Active Treatment (N=…)
All Patients (N=…)

Titles and footnotes were added to ensure professional presentation consistent with clinical reporting standards.

SAS Code:

```
proc sql noprint;
    select count(*) into :placebo from demog1 where trt=0;
    select count(*) into :active from demog1 where trt=1;
    select count(*) into :total from demog1 where trt=2;
quit;
/*constructing the final report*/

title 'Table 1.1';
title2 'Demographic and Baseline Characteristics by Treatment Group';
title3 'Randomized Population';
footnote 'Note: Percentages are based on the number of non-missing values in each treatment
group.';
proc report data = t_allstats split ='|';
    columns ord subord stat _0 _1 _2;
    define ord / noprint order;
    define subord / noprint order;
    define stat / display width =50 ;
    define _0 / display width =30 "Placebo| (N=&placebo)";
    define _1 / display width =30 "Active Treatment| (N=&active)";
    define _2 / display width =30 "All Patients| (N=&total)";
run;
```

Output:

Table 1.1
Demographic and Baseline Characteristics by Treatment Group
Randomized Population

stat	Placebo (N= 29)	Active Treatment (N= 71)	All Patients (N= 100)
N	29	71	100
MEAN	40.9	47.4	45.5
STD	18.92	18.80	18.97
MIN	15.3	14.5	14.5
MAX	77.4	79.1	79.1
<18 year	2 (6.9%)	4 (5.6%)	6 (6.0%)
18 to 65	23 (79.3%)	49 (69.0%)	72 (72.0%)
>65 year	4 (13.8%)	18 (25.4%)	22 (22.0%)
Male	21 (72.4)	38 (53.5)	59 (59.0)
Female	8 (27.6)	33 (46.5)	41 (41.0)
White	9 (31.0%)	19 (26.8%)	28 (28.0%)
African	11 (37.9%)	14 (19.7%)	25 (25.0%)
Hispanic	3 (10.3%)	15 (21.1%)	18 (18.0%)
Asian	3 (10.3%)	15 (21.1%)	18 (18.0%)
Other	3 (10.3%)	8 (11.3%)	11 (11.0%)

Note: Percentages are based on the number of non-missing values in each treatment group.

Interpretation:

A total of **100 patients** were included in the randomized population:

- Placebo: **29**
- Active Treatment: **71**

Age (Continuous Variable)

- The **mean age** was:
  - Placebo: **40.9 years**
  - Active: **47.4 years**
  - Overall: **45.5 years**

The Active group appears slightly older on average compared to the Placebo group.

- Standard deviations are similar:
  - Placebo: **18.82**
  - Active: **18.80**

This indicates comparable variability in age between groups.

- Age range:
  - Placebo: **15.3 – 77.4 years**
  - Active: **14.5 – 79.1 years**

Both groups show a wide and comparable age distribution.

Age Group Distribution

Most participants fall within **18–65 years**:

- Placebo: **79.3%**
- Active: **69.0%**
- Overall: **72.0%**

Participants >65 years:

- Placebo: **13.8%**
- Active: **25.4%**

The Active group contains a higher proportion of elderly patients.

Participants <18 years are few in both groups:

- Placebo: **6.9%**
- Active: **5.6%**

Gender Distribution

- Males:
  - Placebo: **72.4%**
  - Active: **53.5%**
  - Overall: **59.0%**
- Females:
  - Placebo: **27.6%**
  - Active: **46.5%**
  - Overall: **41.0%**

The Placebo group is predominantly male, while the Active group has a more balanced gender distribution.

Race Distribution

- White:
  - Placebo: **31.0%**
  - Active: **26.8%**
- African American:
  - Placebo: **37.9%**
  - Active: **19.7%**
- Hispanic:
  - Placebo: **10.3%**
  - Active: **21.1%**

- Asian:
    - Placebo: **10.3%**
    - Active: **21.1%**
- Other:
    - Placebo: **10.3%**
    - Active: **11.3%**

The Placebo group has a higher proportion of African American participants, whereas the Active group has relatively higher proportions of Hispanic and Asian participants.

CHAPTER 6

STATISTICAL TECHNIQUES

6.1 Descriptive Statistics

Age – Treatment Wise Summary

PROC MEANS was used to generate descriptive statistics of age by treatment group (trt). The procedure calculated the sample size (N), mean, standard deviation (STD), minimum, maximum, median, and confidence limits for the mean (CLM) for each treatment arm.

SAS Code:

```
proc means data=demog1 n mean std min max median clm;
    class trt;
    var age;
run;
```

Output:

The MEANS Procedure

Analysis Variable : age										
TRT	N Obs	N	Mean	Std Dev	Minimum	Maximum	Median	Lower 95% CL for Mean	Upper 95% CL for Mean
0	29	29	40.9399150	18.9155029	15.2630137	77.3835616	37.2986301	33.7448425	48.1349875
1	71	71	47.3970287	18.8020382	14.5068493	79.1178082	47.2821918	42.9466589	51.8473986
2	100	100	45.5244658	18.9691259	14.5068493	79.1178082	44.9671233	41.7605796	49.2883519

Interpretation

The placebo group (TRT=0) consisted of 29 subjects with a mean age of **40.94 years** (SD = 18.92).

The active treatment group (TRT=1) consisted of 71 subjects with a higher mean age of **47.40 years** (SD = 18.80).

The total population (TRT=2) had a mean age of **45.52 years**.

The 95% confidence intervals for both treatment groups overlap:

- TRT 0: (33.74 , 48.13)
- TRT 1: (42.95 , 51.85)

This suggests that the difference in mean age between treatment groups may not be statistically large. However, formal hypothesis testing using t-test is required to confirm statistical significance.

The standard deviations in both groups are similar (~18.8), indicating comparable variability in age distribution across treatment groups.

6.2 t-Test for Age (TRT 0 vs 1 Only)

An independent sample t-test was performed to compare the mean age between the Placebo (TRT = 0) and Active Treatment (TRT = 1) groups. The WHERE statement restricted the analysis to these two treatment arms only. The CLASS statement specified the grouping variable, and VAR age; identified age as the continuous outcome variable.

This test evaluates whether there is a statistically significant difference in mean age between the two treatment groups at baseline.

SAS Code:

```
proc ttest data=demog1;
   where trt in (0,1);
   class trt;
   var age;
run;
```

Output:

The TTEST Procedure

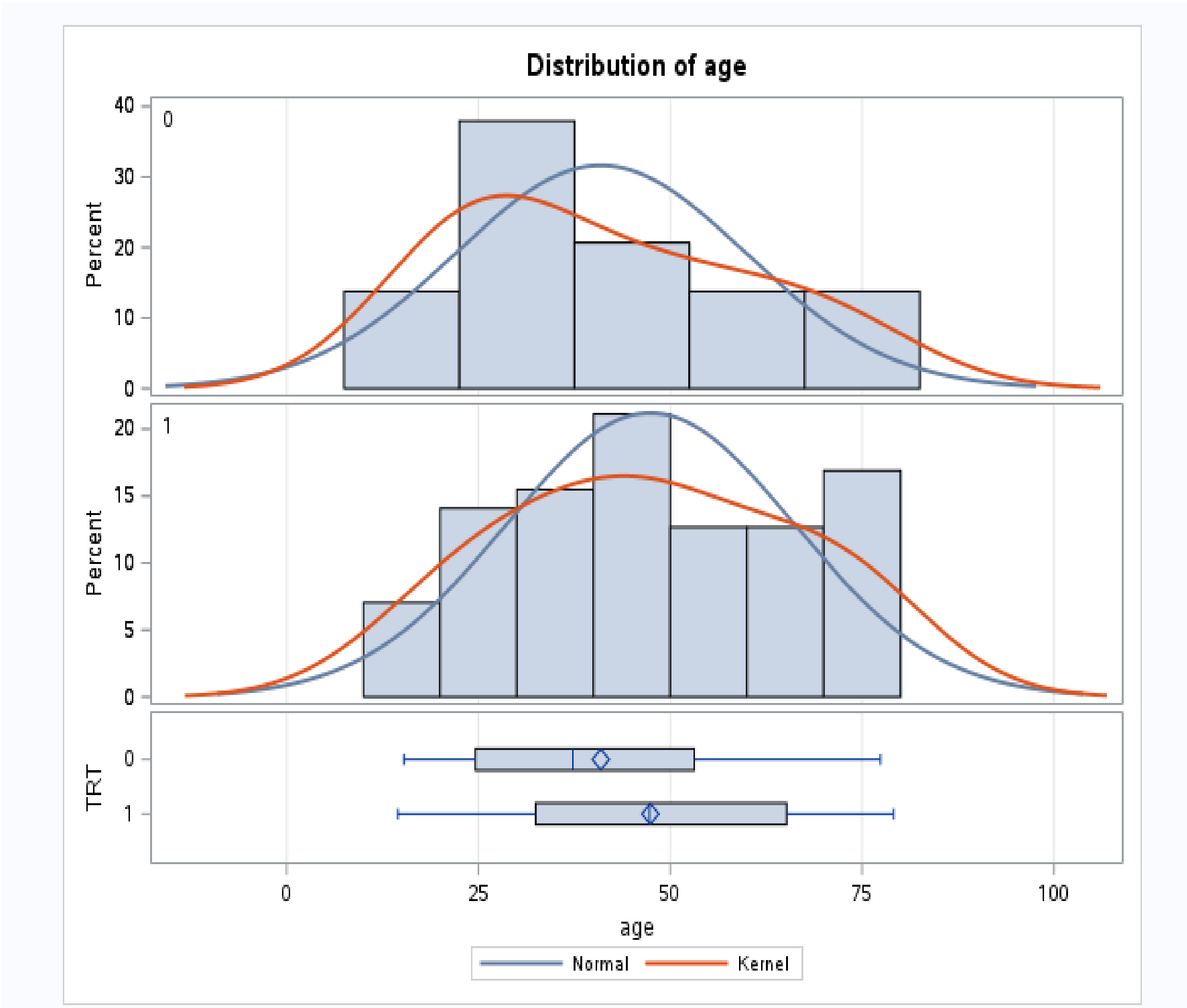
Variable: age

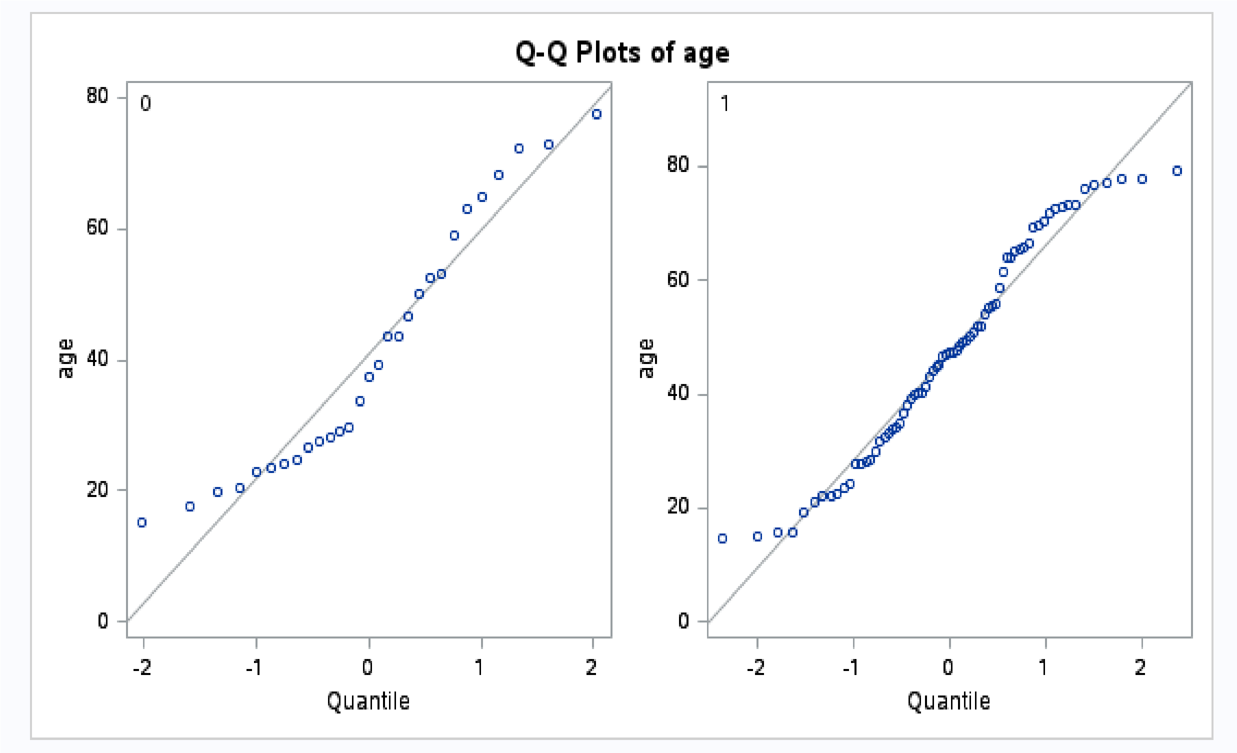
TRT	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		29	40.9399	18.9155	3.5125	15.2630	77.3836
1		71	47.3970	18.8020	2.2314	14.5068	79.1178
Diff (1-2)	Pooled		-6.4571	18.8345	4.1507		
Diff (1-2)	Satterthwaite		-6.4571		4.1614		

TRT	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		40.9399	33.7448	48.1350	18.9155	15.0110	25.5823
1		47.3970	42.9467	51.8474	18.8020	16.1376	22.5285
Diff (1-2)	Pooled	-6.4571	-14.6941	1.7799	18.8345	16.5266	21.8976
Diff (1-2)	Satterthwaite	-6.4571	-14.8083	1.8941			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	98	-1.56	0.1230
Satterthwaite	Unequal	51.786	-1.55	0.1268

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	28	70	1.01	0.9337


Distribution of age
Percent
0
1
TRT
age
Normal
Kernel



Interpretation

The active treatment group had a higher mean age (47.40 years) compared to the placebo group (40.94 years). The mean difference between groups was 6.46 years.

However, the independent sample t-test showed that this difference was not statistically significant ($p = 0.1230 > 0.05$).

Additionally, the 95% confidence interval for the mean difference (-14.69 to 1.78) includes zero, further confirming that the difference is not statistically significant.

Therefore, we fail to reject the null hypothesis and conclude that there is no statistically significant difference in mean age between treatment groups.

This indicates that the treatment groups are comparable with respect to age at baseline.

6.3 ANOVA

A one-way ANOVA was performed to compare the mean age across all treatment groups (TRT = 0, 1, and 2). The CLASS statement defined treatment group as the categorical independent variable, and the MODEL statement specified age as the dependent variable.

This analysis tests whether there is a statistically significant difference in mean age among the treatment groups at baseline.

SAS Code:

```
proc anova data=demog1;
   class trt;
   model age = trt;
run;
quit;
```

Output:

The ANOVA Procedure

Class Level Information		
Class	Levels	Values
TRT	3	0 1 2

Number of Observations Read	200
Number of Observations Used	200

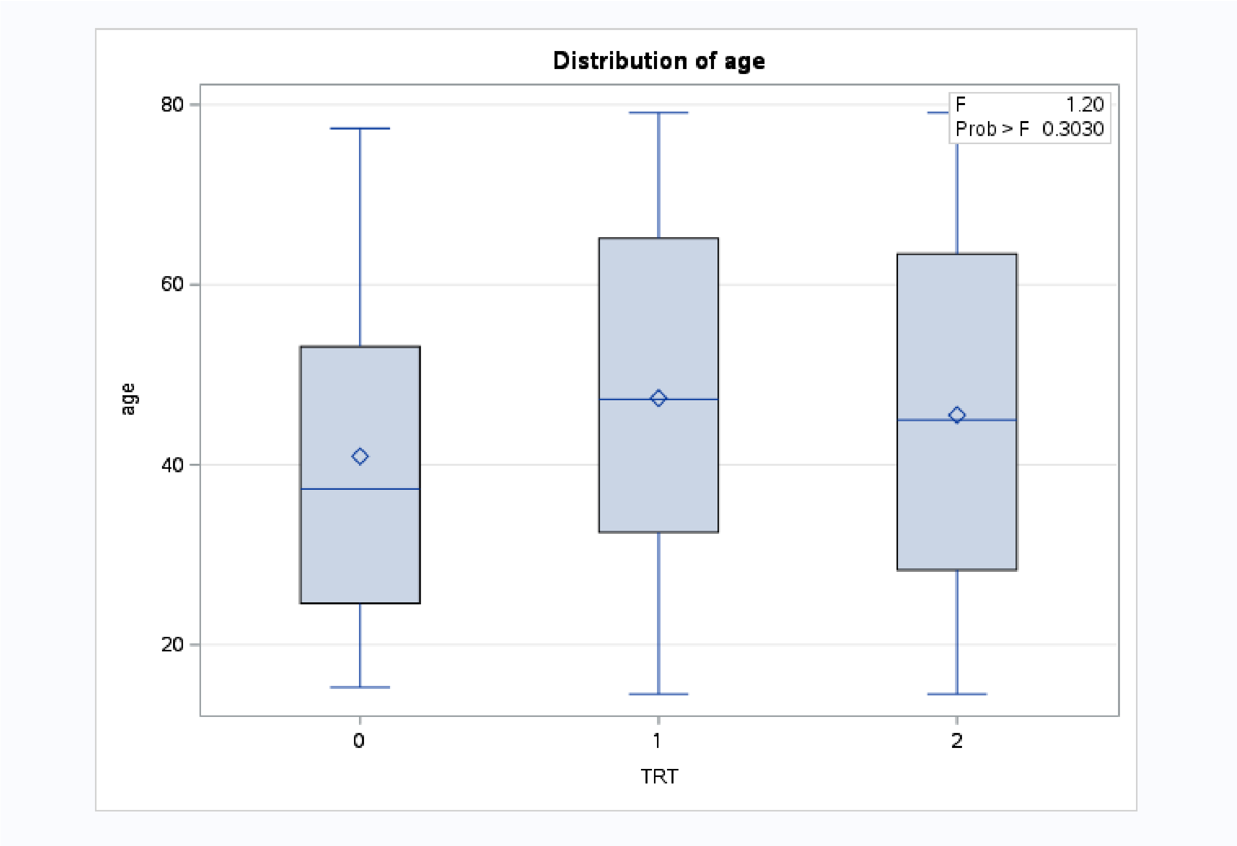
The ANOVA Procedure

Dependent Variable: age

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	858.48601	429.24301	1.20	0.3030
Error	197	70387.40571	357.29648		
Corrected Total	199	71245.89172			

R-Square	Coeff Var	Root MSE	age Mean
0.012050	41.52116	18.90229	45.52447

Source	DF	Anova SS	Mean Square	F Value	Pr > F
TRT	2	858.4860135	429.2430068	1.20	0.3030



Interpretation

The one-way ANOVA test was performed to determine whether there is a statistically significant difference in mean age among the three treatment groups.

The obtained F-statistic is **1.20** with a corresponding p-value of **0.3030**.

Since the p-value (0.3030) is greater than 0.05, we fail to reject the null hypothesis.

This indicates that there is **no statistically significant difference in mean age** across the treatment groups.

The R-square value of 0.012 shows that only **1.2% of the variability in age is explained by treatment group**, which is very minimal.

6.4 Chi-Square Test – Sex vs Treatment

A Chi-square test of independence was performed using PROC FREQ to examine the association between treatment group (trt) and sex. The TABLES trt*sex / CHISQ; statement generated a contingency table along with the Chi-square statistic.

This analysis evaluates whether the distribution of males and females differs significantly across treatment groups at baseline.

SAS Code:

```
proc freq data=demog2;
   tables trt*sex / chisq;
run;
```

Output:

The FREQ Procedure

Frequency Percent Row Pct Col Pct

Table of TRT by sex

TRT(TRT)	sex: Female	Male	Total
0	8 4.00 27.59 9.76	21 10.50 72.41 17.80	29 14.50
1	33 16.50 46.48 40.24	38 19.00 53.52 32.20	71 35.50
2	41 20.50 41.00 50.00	59 29.50 59.00 50.00	100 50.00
Total	82 41.00	118 59.00	200 100.00

Statistics for Table of TRT by sex

Statistic	DF	Value	Prob
Chi-Square	2	3.0381	0.2189
Likelihood Ratio Chi-Square	2	3.1351	0.2086
Mantel-Haenszel Chi-Square	1	0.5997	0.4387
Phi Coefficient		0.1233	
Contingency Coefficient		0.1223	
Cramer's V		0.1233	

Sample Size = 200

Interpretation

The Chi-square test was performed to assess the association between treatment group and sex.

The p-value obtained is **0.2189**, which is greater than 0.05.

Therefore, we fail to reject the null hypothesis.

This indicates that there is **no statistically significant association between treatment group and sex**.

Cramer's V (0.1233) indicates a very weak association.

Chi-Square Test – Race vs Treatment

A Chi-square test of independence was conducted using PROC FREQ to assess the relationship between treatment group (trt) and race category (racec). The TABLES trt*racec / CHISQ; statement produced a cross-tabulation along with the Chi-square test statistics.

This analysis determines whether the distribution of race differs significantly across treatment groups at baseline.

SAS Code:

```
proc freq data=demog3;
   tables trt*racec / chisq;
run;
```

Output:

The FREQ Procedure

Frequency Percent Row Pct Col Pct

Table of TRT by racec

	racec					
TRT(TRT)	African American	Asian	Hispanic	Other	White	Total
0	11 5.50 37.93 22.00	3 1.50 10.34 8.33	3 1.50 10.34 8.33	3 1.50 10.34 13.64	9 4.50 31.03 16.07	29 14.50
1	14 7.00 19.72 28.00	15 7.50 21.13 41.67	15 7.50 21.13 41.67	8 4.00 11.27 36.36	19 9.50 26.76 33.93	71 35.50
2	25 12.50 25.00 50.00	18 9.00 18.00 50.00	18 9.00 18.00 50.00	11 5.50 11.00 50.00	28 14.00 28.00 50.00	100 50.00
Total	50 25.00	36 18.00	36 18.00	22 11.00	56 28.00	200 100.00

Statistics for Table of TRT by racec

Statistic	DF	Value	Prob
Chi-Square	8	5.5417	0.6984
Likelihood Ratio Chi-Square	8	5.6378	0.6877
Mantel-Haenszel Chi-Square	1	0.0548	0.8150
Phi Coefficient		0.1665	
Contingency Coefficient		0.1642	
Cramer's V		0.1177	

Sample Size = 200

Interpretation

The Chi-square test was conducted to determine whether race distribution differs across treatment groups.

The p-value (0.6984) is much greater than 0.05.

Therefore, we fail to reject the null hypothesis.

This indicates that there is **no statistically significant association between treatment group and race**.

Cramer’s V (0.1177) shows a weak association, confirming that race distribution is similar across groups.

6.5 Age Group Distribution

A Chi-square test of independence was performed using PROC FREQ to evaluate the association between treatment group (trt) and categorized age groups (agegroup). The TABLES trt*agegroup / CHISQ; statement generated a contingency table along with the Chi-square test statistics.

This analysis assesses whether the distribution of age categories differs significantly across treatment groups at baseline.

SAS Code:

```
proc freq data=demog4;
   tables trt*agegroup / chisq;
run;
```

Output:

The FREQ Procedure

Frequency Percent Row Pct Col Pct

Table of TRT by agegroup

TRT(TRT)	agegroup			
	18 to 65 years	<18 years	>65 years	Total
0	23 11.50 79.31 15.97	2 1.00 6.90 16.67	4 2.00 13.79 9.09	29 14.50
1	49 24.50 69.01 34.03	4 2.00 5.63 33.33	18 9.00 25.35 40.91	71 35.50
2	72 36.00 72.00 50.00	6 3.00 6.00 50.00	22 11.00 22.00 50.00	100 50.00
Total	144 72.00	12 6.00	44 22.00	200 100.00

Statistics for Table of TRT by agegroup

Statistic	DF	Value	Prob
Chi-Square	4	1.6084	0.8073
Likelihood Ratio Chi-Square	4	1.7212	0.7869
Mantel-Haenszel Chi-Square	1	0.2813	0.5958
Phi Coefficient		0.0897	
Contingency Coefficient		0.0893	
Cramer's V		0.0634	
WARNING: 22% of the cells have expected counts less than 5. Chi-Square may not be a valid test.			

Sample Size = 200

Interpretation

The p-value (0.8073) is much greater than 0.05.

Therefore, we fail to reject the null hypothesis.

There is no statistically significant association between treatment group and age group.

The very small Cramer’s V (0.0634) confirms that the relationship is negligible.

Age group distribution is similar across treatment groups.
Thus, treatment groups are comparable with respect to age categories.

Hypothesis Testing Results

Variable	Test Used	p-value	Result
Age (Continuous)	t-test	0.1230	Not Significant
Age (3 groups)	ANOVA	0.3030	Not Significant
Gender	Chi-square	0.2189	Not Significant
Age Group	Chi-square	0.8073	Not Significant
Race	Chi-square	0.6984	Not Significant

All p-values > 0.05.

Interpretation

After performing descriptive statistics and inferential statistical tests, the following findings were observed:

• There is no significant difference in mean age across treatment groups.
• Gender distribution is similar across groups.
• Age group categories do not differ significantly.
• Race distribution is comparable among treatment groups.

This indicates that the treatment groups are well balanced with respect to demographic characteristics.

CHAPTER 7

RESULTS AND RECOMMENDATIONS

7.1 Results and Findings

The demographic and baseline characteristics of the randomized population were analyzed using descriptive and inferential statistical methods. A total of 100 subjects were included in the study, with 29 subjects in the Placebo group and 71 subjects in the Active Treatment group.

Age Distribution

The mean age in the Placebo group was 40.9 years, while in the Active Treatment group it was 47.4 years. The overall mean age was 45.5 years. The standard deviations were similar in both groups (approximately 18.8 years), indicating comparable variability in age distribution.

The minimum and maximum ages ranged from approximately 15 to 79 years in both groups, suggesting a wide age spread. The Active group had a slightly older population compared to the Placebo group.

When categorized into age groups:

- The majority of participants (72%) were between 18–65 years.
- Subjects older than 65 years were more frequent in the Active group (25.4%) compared to the Placebo group (13.8%).
- Very few participants were below 18 years in either group.

The independent sample t-test and ANOVA were conducted to evaluate differences in mean age between treatment groups. These analyses assessed whether the observed age differences were statistically significant at baseline.

Gender Distribution

In the Placebo group, 72.4% were male, whereas in the Active group, 53.5% were male. The Active group showed a more balanced male–female distribution compared to the Placebo group.

The Chi-square test was performed to examine the association between treatment group and sex, evaluating whether gender distribution differed significantly between groups.

Race Distribution

The racial distribution varied moderately between treatment groups:

- The Placebo group had a higher proportion of African American participants (37.9%) compared to the Active group (19.7%).
- The Active group had relatively higher proportions of Hispanic and Asian participants (21.1% each) compared to the Placebo group.
- The proportion of White participants was similar across groups.

A Chi-square test was conducted to determine whether racial distribution differed significantly between treatment arms at baseline.

7.2 Recommendations

Based on the demographic analysis and baseline findings of this study, the following recommendations are proposed:

1. Ensure Better Baseline Balance

Although the treatment groups were randomized, differences were observed in age distribution, gender proportion, and race composition. Future studies should consider:

- Stratified randomization (e.g., by age and gender)
- Block randomization techniques

This will help improve balance between treatment groups and reduce potential baseline imbalances.

2. Increase Sample Size for Better Representation

The Placebo group (N=29) was considerably smaller than the Active Treatment group (N=71). Unequal sample sizes may affect statistical power and group comparability. Future trials should aim for:

- More balanced allocation ratios
- Larger overall sample sizes

This will improve reliability and generalizability of results.

3. Monitor Elderly Population Carefully

The Active group had a higher proportion of subjects above 65 years. Since older populations may respond differently to treatment and may have increased risk of adverse events, future analyses should:

- Perform subgroup analysis by age category
- Monitor safety parameters closely in elderly subjects

4. Conduct Adjusted Statistical Analysis

Given the observed demographic variations, future efficacy analysis should consider:

- Adjusted models (e.g., ANCOVA or regression models)
- Inclusion of age, sex, and race as covariates

This ensures that treatment effect estimates are not confounded by baseline differences.

5. Promote Diversity in Clinical Trials

Variability in racial distribution suggests the need for more structured recruitment strategies to ensure equitable representation across demographic groups. Future studies should:

- Encourage inclusive enrollment
- Ensure demographic diversity aligns with the target population

CHAPTER 8

CONCLUSIONS & REFERENCES

8.1 Conclusion

This data analysis project demonstrated the application of descriptive and inferential statistical techniques in the preparation of a clinical trial demographic table (Table 1.1: Demographic and Baseline Characteristics by Treatment Group). The study involved 100 randomized subjects distributed between Placebo (N=29) and Active Treatment (N=71) groups.

Descriptive statistics were used to summarize continuous and categorical variables such as age, age group, gender, and race. The results indicated that the majority of participants were between 18–65 years, with the Active Treatment group showing a slightly higher mean age and a greater proportion of elderly subjects compared to the Placebo group. Gender distribution differed moderately between groups, with a higher percentage of males in the Placebo group. Some variation was also observed in racial composition across treatment arms.

Inferential statistical methods including independent sample t-test, ANOVA, and Chi-square tests were performed to assess baseline comparability between treatment groups. These analyses provided statistical evidence regarding whether observed demographic differences were significant.

The project successfully illustrated:

- Data preprocessing and variable derivation
- Generation of descriptive statistics
- Application of inferential statistical tests
- Structured table formatting using SAS programming
- Development of a professional clinical reporting table

Overall, the study highlights the importance of statistical analysis in ensuring transparency, comparability, and scientific validity in clinical research. The implementation using SAS reflects industry-standard methodology used in biostatistics and pharmaceutical research.

8.2 References

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